A Decoupled Dual-Teacher Collaborative Semisupervised Framework for High-Resolution Remote Sensing Image Semantic Segmentation

Ting Zhang, Chenxu Ge, Qiangkui Leng, *Member, IEEE*, Hisham Alfuhaid, *Member, IEEE*, Mohammad Al-Antary, Muhammad Waqas, *Senior Member, IEEE*, and Abdallah Namoun, *Senior Member, IEEE*

***Abstract*—Semantic segmentation of high-resolution remote sensing images is a fundamental task in Earth observation, yet its performance is often constrained by the expensive and time-consuming process of pixel-level annotation. While existing semisupervised learning (SSL) methods can mitigate this label scarcity, they often overlook the inherent spectral–spatial duality of remote sensing data. This oversight leads to the entanglement of heterogeneous features and the progressive accumulation of pseudolabel noise, ultimately limiting segmentation accuracy. To address this, we propose a decoupled dual-teacher collaborative semisupervised framework (DDCSF), which generates high-quality supervisory signals from heterogeneous modalities via novel decoupling and alignment mechanisms. DDCSF employs separate spectral and spatial teacher networks to achieve decoupled learning of texture features and geometric structures, respectively. To align these heterogeneous features, we design a frequency-aware cross-modal fusion module (FCFM). This module leverages the high-frequency components of spatial features to refine target boundaries and the low-frequency components to correct regional semantics, producing collaborative features with both precise details and semantic consistency. Furthermore, a pixel confidence voting module (PCVM) is introduced to quantify prediction uncertainty from the dual teachers and fusion module, selecting high-confidence pseudolabels to guide student model training. The student model's ability to represent multimodal inputs is further enhanced by an adaptive spectral–spatial image fusion module (ASSIFM). Experimental results on the ISPRS Potsdam and GID benchmark datasets demonstrate that DDCSF consistently outperforms state-of-the-art methods at various label rates, validating its effectiveness.**

***Index Terms*—Dual-teacher network, frequency-domain perception, semisupervised semantic segmentation, spectral–spatial synergy.**

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Ting Zhang is with the College of Computer Science, Beijing University of Technology, Beijing 100124, China (e-mail: zhangting@bjut.edu.cn).

Chenxu Ge and Qiangkui Leng are with the School of Electronic and Information Engineering, Liaoning Technical University, Huludao 125105, China (e-mail: 18803131235@163.com; qkleng@126.com).

Hisham Alfuhaid is with the Department of Computer Science, King Khalid University, Abha 61421, Saudi Arabia (e-mail: alasmary@kku.edu.sa).

Mohammad Al-Antary and Muhammad Waqas are with the School of Computing and Mathematical Science, Faculty of Engineering and Science, University of Greenwich, SE10 9LS London, U.K. (e-mail: m.t.m.alantary@greenwich.ac.uk; engr.waqas2079@gmail.com).

Abdallah Namoun is with AI Center, Islamic University of Madinah, Madinah 42351, Saudi Arabia, and also with the Faculty of Computer and Information Systems, Islamic University of Madinah, Madinah 42351, Saudi Arabia (e-mail: a.namoun@iu.edu.sa).

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I. Introduction

HIGH resolution semantic segmentation of remote sensing images, which assigns predefined meaning labels to each pixel, is a cornerstone technology for applications in Earth observation, including urban dynamic monitoring and evaluation [1], resource management [2], and disaster response [3]. The advent of deep learning has led to significant progress, with fully supervised segmentation networks [4] demonstrating powerful feature representation capabilities. However, the performance of these models is heavily reliant on large-scale datasets with fine-grained pixel-level annotations. In the remote sensing domain, obtaining such high-precision pixel labels is a laborious and costly process that demands significant domain expertise. This annotation bottleneck severely hinders the development of advanced segmentation models in dynamic real-world scenarios. Consequently, semisupervised learning (SSL), which leverages the vast amount of unlabeled data to improve model performance, has emerged as a critical solution [5], [6], [7].

Current semisupervised research is dominated by self-training frameworks based on consistency regularization and pseudolabeling. Classic teacher–student paradigms enforce prediction consistency on unlabeled data under perturbations, generating pseudolabels to guide training [8], [9]. While these methods have achieved considerable success on natural image datasets like cityscapes and Pascal VOC, their direct application to remote sensing images often yields suboptimal results. Although several SSL methods have been developed specifically for remote sensing images [10], [11], they typically make only partial use of the unique characteristics of remote sensing targets. Their core philosophy still treats these images as standard RGB inputs, failing to adequately account for the inherent spectral–spatial duality of the data. More

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critically, under weak supervision constraints, such methods suffer from severe modal collapse; the network tends to neglect the minority feature modality while overrelying on the dominant one, giving rise to systematic errors that further degrade segmentation performance [12], [13]. Remote sensing imagery contains two distinct yet complementary information streams: spectral information, which helps distinguish the physical properties of different materials (e.g., differentiating water from shadows or asphalt from cement), and spatial information, which defines the geometric boundaries and texture structure of ground objects [14], [15], [16]. Most existing SSL methods employ simple early fusion strategies, such as direct channel stacking at the input data level. This coarse approach introduces two primary problems

1) *Feature Entanglement and Semantic Ambiguity:* The spectral signatures of different land cover types often lie on distinct manifolds. Forcibly fusing these without a principled decoupling mechanism can cause the model to favor the dominant material properties of one modality while suppressing crucial cues from the other. This imbalance can lead to semantic ambiguity, where spectrally similar but unrelated objects are confused, resulting in blurred or inaccurate predictions in complex scenes.
2) *Modal Collapse Induced Pseudolabel Noise and Confirmation Bias:* The fundamental issue is not merely random pseudolabel noise, but modal collapse under weak supervision. The network prioritizes dominant feature modalities while ignoring the less prominent ones, causing systematic prediction errors. Consequently, teacher-generated pseudolabels are prone to heavy noise near object boundaries and missing detections of small objects. Iterative learning from such biased pseudolabels further leads to irreversible confirmation bias, severely limiting segmentation performance.

To enhance the accuracy of semisupervised remote sensing image segmentation, we present a decoupled dual-teacher collaborative semisupervised framework (DDCSF). DDCSF adopts a divide-and-conquer strategy by constructing two separate teacher networks: a spectral teacher focused on extracting material properties from the spectral domain, and a spatial teacher dedicated to capturing geometric structures from the spatial domain. As illustrated in Fig. 1, different land cover types exhibit unique responses in the spectral dimension based on their spatial distribution, material properties, and temporal structure. Models relying solely on either spatial or spectral information will inevitably encounter ambiguity caused by spectrally similar but structurally distinct objects. This limitation is a primary source of low-confidence pseudolabels in areas with complex boundaries, small targets, and spectral confusion [17], [18], [19].

To address these challenges from a frequency-domain perspective, we propose three novel modules. First, an adaptive spectral–spatial image fusion module (ASSIFM) adaptively weights multimodal information at the input stage to reduce the knowledge gap for the student model. Second, a frequency-aware cross-modal fusion module (FCFM) leverages the high-frequency components of spatial features to guide the precise refinement of spectral feature boundaries while using low-frequency components to align regional semantics. This approach yields high-quality fused features with both accurate representations and complete geometric structures. To suppress pseudolabel noise and mitigate confirmation bias, we also introduce a pixel confidence voting module (PCVM), which fuses multisource predictions. By quantifying pixel-level uncertainty and employing a voting mechanism, PCVM selects the most reliable pseudolabels for training.

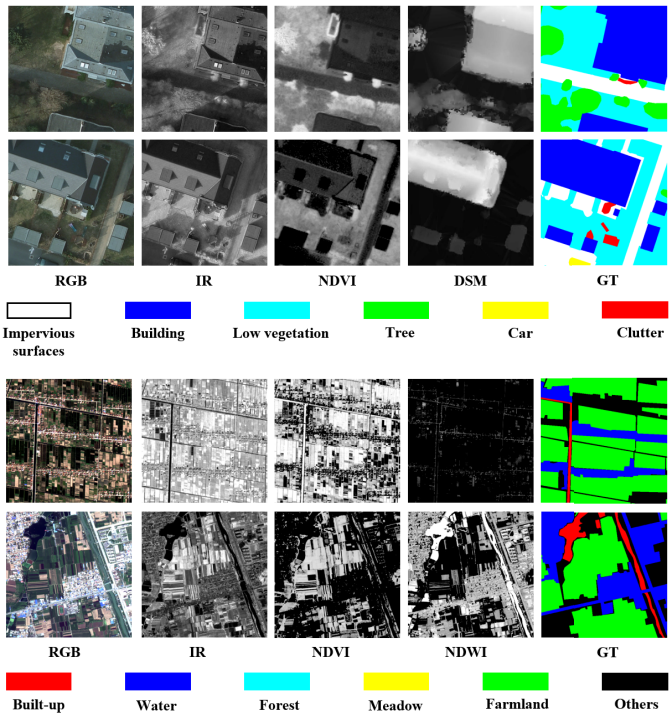


Fig. 1. Spatial distribution, material properties, and periodic structures of different land-cover types exhibit distinctive representations in the spectral domain.

The main contributions are summarized as follows.

1) We propose a DDCSF. It tackles low-quality pseudolabels from remote sensing modality heterogeneity via independent modeling and deep collaboration.
2) We construct an ASSIFM to adaptively fuse raw spectral and spatial images, providing the student model with more informative and enriched inputs.
3) We design a FCFM that introduces frequency-domain decoupling. The high- and low-frequency components of spatial features, respectively, guide boundary refinement and semantic alignment of spectral features, thereby enhancing the discriminative power of feature fusion.
4) We develop a PCVM. It adopts pixelwise confidence filtering and multisource voting. It maximizes unlabeled data utilization.
5) Experimental results on the ISPRS Potsdam and GID benchmark datasets demonstrate that DDCSF consistently achieves superior segmentation accuracy over current state-of-the-art methods under various label ratio settings, validating the effectiveness and superiority of our proposed approach.

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II. Related Work

Our work is mainly related to three research areas: remote sensing image semantic segmentation, SSL methods, and multisource information fusion techniques.

A. High-Resolution Remote Sensing Image Semantic Segmentation

Since the advent of deep learning, end-to-end models represented by fully convolutional networks (FCNs) [20], [21], [22] have become the dominant paradigm. Research to improve segmentation accuracy in remote sensing has largely focused on two directions. The first is multiscale context modeling, which captures land-cover information at different scales by incorporating dilated convolutions or spatial pyramid structures, as seen in PSPNet [23], DeepLabv3 + [24], and ABCNet [25]. The second direction is boundary refinement and feature enhancement, which leverages attention mechanisms (e.g., DANet [26], nonlocal Net [27], CMTFNet [28]) or uses high-level semantic information to guide the recovery of low-level features and preserve spatial details. Recently, Transformers and their variants have been successfully applied to computer vision, and their powerful global dependence modeling capability has been effectively exploited for remote sensing image segmentation [29], [30], [31], [32]. For instance, SegFormer [33] and Swin Transformer [34] employ self-attention mechanisms to capture long-range spatial dependencies, demonstrating significant advantages in processing large-scale remote sensing images. DC-Swin [35] further enhances multiscale feature fusion with a densely connected decoder.

Despite their promising performance, these fully supervised methods share two common limitations. First, they heavily rely on large volumes of meticulously annotated data. Second, most adopt single-stream network architectures that treat multispectral channels as a simple input concatenation. Although some methods use dual-branch structures to process visible and near-infrared (NIR) information separately, their fusion strategies are often coarse (e.g., channelwise concatenation or elementwise summation) [36], [37], [38]. While these models can fit complex features under strong supervision, in data-scarce semisupervised scenarios, they lack mechanisms for robust decoupling and collaboration of heterogeneous modal features. This deficiency results in unstable and unreliable predictions on unlabeled data [39], [40].

B. Semisupervised Remote Sensing Image Semantic Segmentation

Semisupervised semantic segmentation aims to improve model performance by leveraging a small amount of labeled data together with a large volume of unlabeled data. Mainstream methods can be broadly categorized into two classes: consistency regularization-based methods and pseudolabel-based methods. Consistency regularization-based methods apply different perturbations (e.g., data augmentation or network parameter perturbations) to unlabeled data and constrain the model to produce consistent predictions. For example, mean teacher [41], [42] constructs a teacher model via an exponential moving average of student parameters, improving prediction stability and serving as a foundation for subsequent research. UniMatch [43] extends this by introducing strong–weak augmentation consistency constraints and enhancing discriminability through feature-level contrastive learning. Although effective, the performance of these methods is highly dependent on the design of perturbations, and their robustness can be limited when facing the complex spectral and structural variations in remote sensing imagery.

Pseudolabel-based methods use teacher models to generate high-confidence pseudolabels for unlabeled data, thereby expanding the training set. FixMatch [44], for instance, combines weak and strong augmentations to generate pseudolabels and enforce consistency, with its effectiveness largely depending on the choice of a confidence threshold. CPS [45] mitigates single-model bias through cross-supervision between dual networks. MUCA [46] introduces multiscale uncertainty estimation to improve pseudolabel quality near object boundaries. DWL [47] reduces the impact of erroneous pseudolabels through decoupled learning modules and mitigates majority-class bias via ranking-based weighting. SGCT-Net [48] leverages the contour perception capability of SAM to refine coarse pseudolabels generated by class activation maps (CAMs) and alleviate the fragmentation of ground object edges. ATSG [49] constructs an adaptive token linking module to mitigate the oversmoothing issue of ViT models in dense ground feature scenarios, and embeds a SAM-based boundary correction unit to rectify misclassified regions within pseudolabels, significantly reducing intraclass confusion. The adaptive Gaussian mixture model [50] conducts weak-supervised modeling via Gaussian mixture distributions, mines semantic correlations between pixels from the feature space, and adaptively calibrates low-confidence pseudolabel regions. However, most of these methods are developed from a single-model perspective and fail to cross-validate or enhance pseudolabel reliability from the viewpoint of multisource, heterogeneous information [51].

C. Multisource Information Fusion in Remote Sensing

The remote sensing community has long focused on fusing multisource heterogeneous data to improve interpretation accuracy. Indeed, remote sensing image analysis is inherently a multisource information processing task [52], [53], [54]. While traditional fusion methods operated at the pixel level, the deep learning era has shifted the focus to feature-level fusion. Simple operations like channel concatenation or elementwise summation are computationally efficient [55], [56] but struggle to adaptively regulate the relative importance of different modalities across varying scenes. To achieve deeper feature fusion, various attention-driven modules have been proposed. For example, the spatial–spectral fusion BiFormer (SSFBF) [57] combines the local perception of CNNs with the global attention of Transformers. Spatial–spectral mixed attention (SSMA) [58] designs dedicated attention maps to recalibrate modal features, and the deep fusion module (DFM) [59] focuses on multiscale feature integration.

To address the limitations of the single spatial domain, some researchers have begun to explore integrating

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frequency-domain information into segmentation networks, as frequency-domain features are crucial for capturing textures, edges, and periodic patterns in remote sensing images [60], [61], [62]. For instance, FreDNet [63] employs the DRB module to suppress frequency-domain noise and preserve edge structures, improving the frequency robustness of segmentation predictions. FSSFFormer [64] first decomposes remote sensing images into low- and high-frequency subbands using a fixed 2-D discrete wavelet transform (DWT) to retain global spatial structures and fine details, then refines features via the SEA and SSCA modules. FDNet [65] enhances feature representations by independently refining frequency-domain high- and low-frequency components, and achieves superior segmentation accuracy by optimizing feature learning with three core modules: sparse-aware spectrum enhancement, frequency decoupling attention, and attentive frequency context. However, existing approaches still suffer from two limitations. First, most fusion modules operate exclusively in the spatial domain, overlooking the inherent advantages of frequency-domain information in separating textures from structures. This leads to insufficiently refined boundary details in the fused features. Second, current semisupervised frameworks typically treat fusion modules as internal network components rather than as a core mechanism for improving pseudolabel quality [66], [67], [68].

To address these issues, we propose a frequency-aware cross-modal fusion mechanism. It explicitly exploits the high-frequency components of spatial features to guide boundary refinement of spectral features while using low-frequency components to achieve semantic alignment. This design moves beyond simple feature aggregation, instead employing a physics-inspired decoupling and reconstruction process to provide more accurate and robust supervisory signals for SSL.

III. Method

This section provides a detailed description of the proposed DDCSF. First, the overall architecture of the framework is outlined. Then, its core components are introduced in detail, including the decoupled dual-teacher architecture, the FCFM, and the PCVM. Finally, the loss functions of the entire framework are presented.

A. Overall Architecture

DDCSF is designed to address the challenges of spectral–spatial feature entanglement and pseudolabel noise accumulation in high-resolution remote sensing image segmentation. The framework adopts a dual-teacher–single-student architecture, where the two teacher networks focus on the decoupled feature extraction of spectral and spatial modalities, respectively, thereby avoiding the feature confusion inherent in traditional single-stream networks. The student network learns a fused representation from the dual teachers through a collaborative mechanism. The overall architecture is illustrated in Fig. 2. As shown in Fig. 2, the input remote sensing images first undergo initial modality decoupling. Subsequently, the dual teacher networks independently encode spectral and spatial features. These features are then processed by the FCFM to generate a collaborative prediction. The predictions from both teachers and the FCFM are integrated by the PCVM to select the most reliable hybrid pseudolabels. Meanwhile, the ASSIFM adaptively fuses the original images to create an enhanced input for the student model. Finally, during semisupervised training, the model parameters are jointly optimized using consistency and pseudolabel losses, enabling the effective exploitation of unlabeled data.

The entire training process is divided into two stages.

1) *High-Quality Pseudolabel Generation:* For each unlabeled data point, the spectral and spatial teachers perform independent inference from their respective modality perspectives. The initial predictions produced by both teachers, along with a third-party collaborative prediction generated by the FCFM and an auxiliary teacher decoder, are fed into the PCVM. The PCVM conducts pixelwise uncertainty estimation and selects the most confident predictions to serve as the final pseudolabels.
2) *Student Model Training:* The high-quality pseudolabels generated in the first stage act as supervisory signals for the unlabeled data. These are used in conjunction with the ground-truth labels from the labeled data to guide the training of the student model. To further enhance the student's multimodal representation capability, the ASSIFM provides an adaptively fused image as a more informative input.

FCFM, ASSIFM, and PCVM are our three core modules. They work in concert to form a comprehensive system. The system realizes high-quality pseudolabel generation and learning. This finally improves segmentation accuracy.

B. Dual-Teacher Encoder Architecture

To achieve deep decoupling of spectral and spatial information, we abandon the single-teacher paradigm and instead design two teacher networks with identical architectures but distinct inputs and learning objectives.

Spectral Teacher: It receives augmented multispectral imagery as input and is designed to learn the material attributes and semantic class information of ground objects.

Spatial Teacher: It receives an augmented spatial-structure image as input and aims to capture the geometric boundaries, shapes, and spatial context of objects while remaining insensitive to spectral variations.

1) Adaptive Spectral–Spatial Image Fusion Module: To enable the student model to more effectively learn the complementarity between spectral and spatial information, ASSIFM performs adaptive weighted fusion at the input level instead of simple channel stacking. The core idea is to dynamically learn the weights for the spectral and spatial inputs, allowing the model to adaptively emphasize the modality that contains more informative cues according to the current scene. In this way, ASSIFM provides the student model with an optimally preprocessed input where information is enhanced, and modalities are preliminarily aligned.

As illustrated in Fig. 3, this module first uses a shared encoding layer to extract the corresponding encoded features

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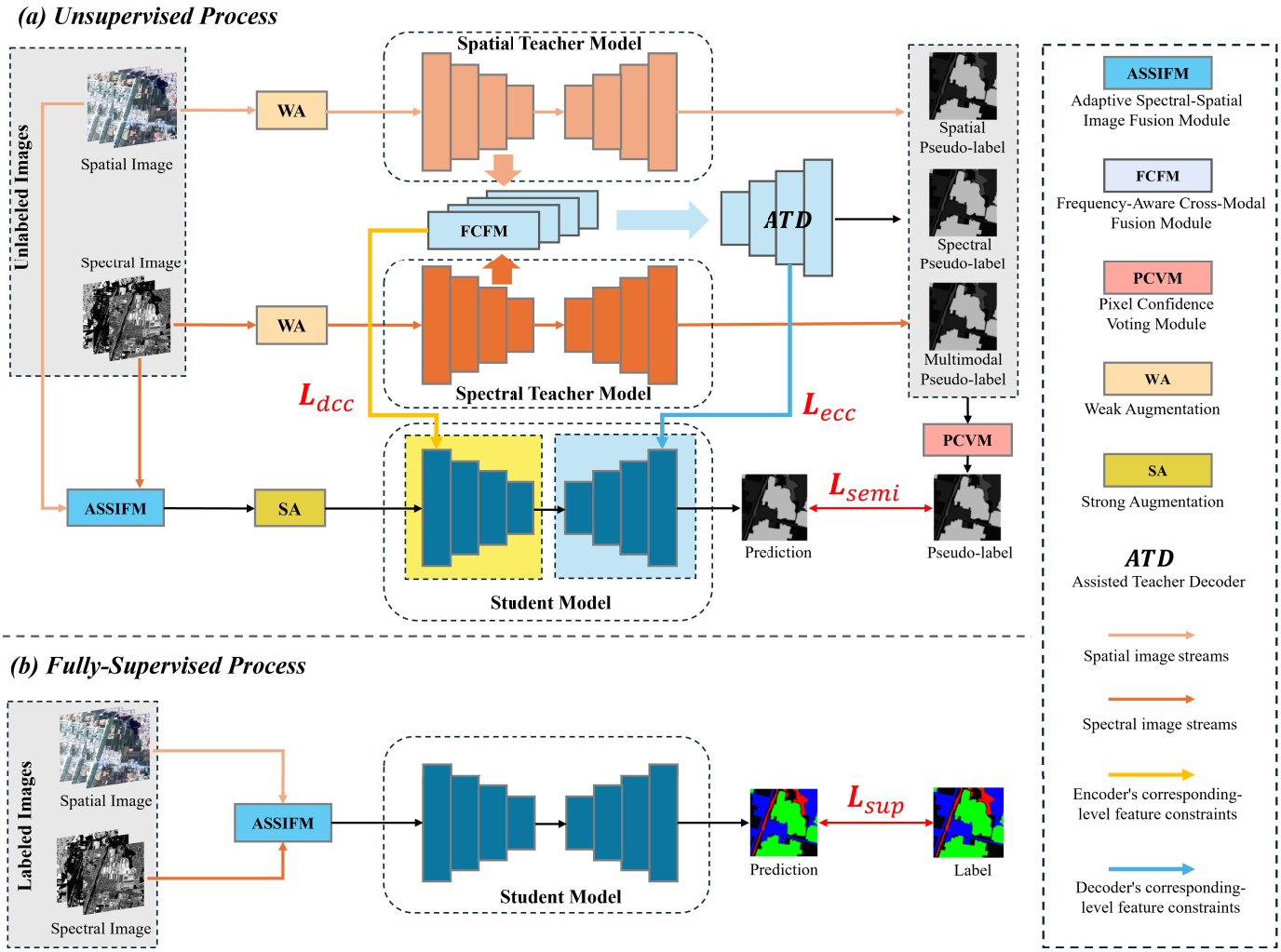


Fig. 2. Overall architecture of DDCSF. Built upon a dual-teacher–student framework, DDCSF introduces a decoupled design with spectral and spatial teachers. High-fidelity pseudolabels are generated through the FCFM and the PCVM, while an adaptive input fusion module and multigranularity consistency constraints (MGCCs) are employed to guide efficient learning of the student model. (a) Unsupervised process. (b) Fully supervised process.

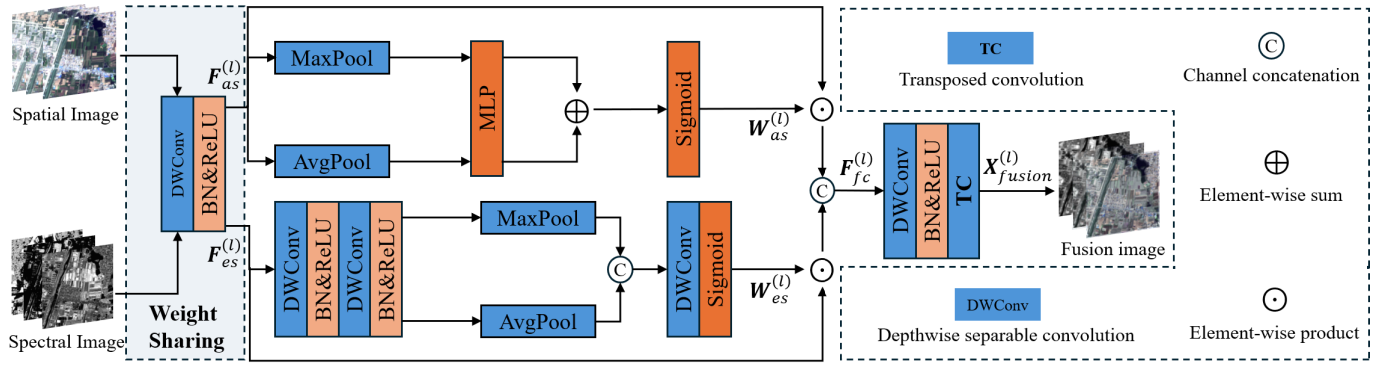


Fig. 3. ASSIFM. This module dynamically generates fusion weights through a channel-attention mechanism, enabling adaptive enhancement of spectral and spatial inputs.

$F_{as}^{(l)}$ and $F_{es}^{(l)}$ from the spatial and spectral inputs, respectively. Global max pooling and global average pooling are then applied to capture channelwise global information. Subsequently, a multilayer perceptron (MLP) followed by a sigmoid activation function is used to generate the spatial channel weight map $W_{as}^{(l)}$ and the spectral channel weight map $W_{es}^{(l)}$. Then, the weight maps $W_{as}^{(l)}$ and $W_{es}^{(l)}$ are multiplied with the original spatial encoded feature $F_{as}^{(l)}$ and spectral encoded feature $F_{es}^{(l)}$, respectively, followed by concatenation to obtain the preliminary fused feature $F_{fc}^{(l)}$. The final fused image $X_{fusion}^{(l)}$ is obtained by a weighted summation of the spectral and spatial inputs using their respective weight maps. This

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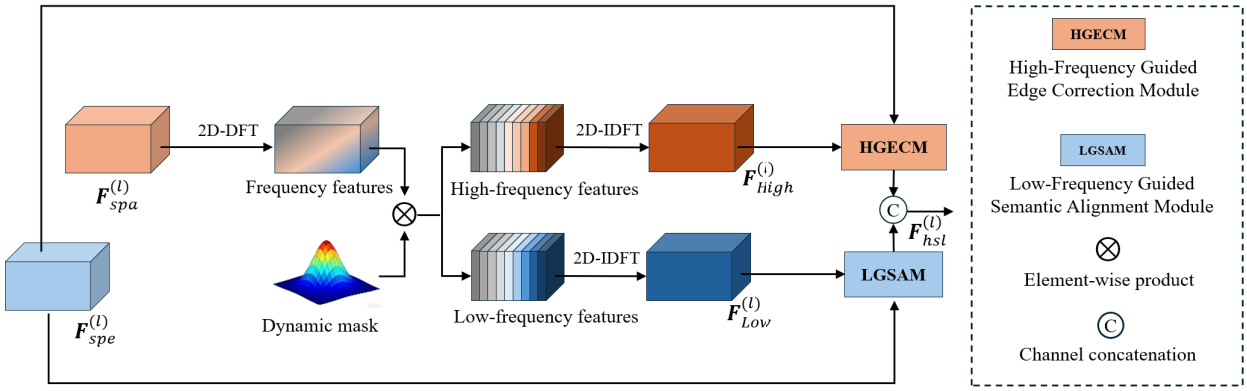


Fig. 4. Overall architecture of the FCFM.

process can be formulated as

$$\boldsymbol{F}_{fc}^{(l)} = Cat\left(\boldsymbol{F}_{as}^{(l)} \odot \boldsymbol{W}_{as}^{(l)}, \boldsymbol{F}_{es}^{(l)} \odot \boldsymbol{W}_{es}^{(l)}\right) \tag{1}$$

$$\boldsymbol{X}_{fusion}^{(l)} = TC\left(ReLU\left(BN\left(DWConv\left(\boldsymbol{F}_{fc}^{(l)}\right)\right)\right)\right) \tag{2}$$

where $\boldsymbol{F}_{es}^{(l)}$ and $\boldsymbol{F}_{as}^{(l)}$ denote the spectral and spatial encoded features, respectively; $\boldsymbol{F}_{fc}^{(l)}$ denote the preliminary fused feature, $TC(\cdot)$ represents the transposed convolution operation, and $\odot$ denotes elementwise multiplication.

C. Frequency-Aware Cross-Modal Fusion Module

To overcome the limitations of single-modality features and synthesize high-quality collaborative features for pseudolabel generation, we propose the FCFM module. Grounded in the physical principle of remote sensing imaging, where high frequencies correspond to fine details (edges and textures) and low frequencies encode global contours/contextual information—and tailored to the intrinsic properties of CNNs, which feature weak global modeling and inefficient frequency feature extraction, FCFM adopts frequency-domain decoupling. It decomposes spatial features into high- and low-frequency components, which respectively guide the boundary refinement and semantic alignment of spectral features. The architecture of FCFM is shown in Fig. 4.

In particular, the spatial features extracted by the spatial teacher are first transformed into the frequency domain using a 2-D discrete Fourier transform (2D-DFT). Dynamic masks are then applied to separate the frequency-domain representation into high-frequency and low-frequency components. Notably, we adopt 0.2 as the initial low-frequency cutoff ratio, which aligns with the spatial–spectral feature distribution and adapts to different feature map scales. Meanwhile, a dynamic mask with fixed shape and learnable filtering strength is employed, which avoids spectral distortion, preserves the physical meaning of frequency decomposition, and enables adaptive optimization through end-to-end training to improve model generalization. Subsequently, a 2-D inverse discrete Fourier transform (2D-IDFT) projects these components back into the spatial domain. The high-frequency components are used to guide boundary refinement of the spectral features, while the low-frequency components facilitate semantic alignment. The enhanced spectral features are then fused to produce the final collaborative features. This process can be formulated as

$$M(i,j) = \begin{cases} 1, & \sqrt{(i - H/2)^2 + (j - W/2)^2} \le \dfrac{\min(H,W)\cdot\gamma}{2} \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

$$M_{dyn} = \sigma(\theta)\cdot M + (1 - \sigma(\theta))\cdot(1 - M) \tag{4}$$

$$\boldsymbol{F}_{high}^{(l)} = \mathrm{Re}\left(\mathcal{F}^{-1}\left(\left|\mathcal{F}\left(\boldsymbol{F}_{spa}^{(l)}\right)\right| \odot \left(1 - M_{dyn}\right)\cdot e^{j\angle\mathcal{F}(\boldsymbol{x})}\right)\right) \tag{5}$$

$$\boldsymbol{F}_{low}^{(l)} = \mathrm{Re}\left(\mathcal{F}^{-1}\left(\left|\mathcal{F}\left(\boldsymbol{F}_{spa}^{(l)}\right)\right| \odot \left(M_{dyn}\cdot e^{j\angle\mathcal{F}(\boldsymbol{x})}\right)\right)\right) \tag{6}$$

where $M(i,j)$ denotes the fixed circular low-pass frequency mask; (i,j) are spectrum pixel coordinates; H, W are the height and width of the input feature map; and γ is the low-frequency cutoff ratio; M_{dyn} denotes the learnable dynamic frequency mask; $\sigma(\cdot)$ is the sigmoid activation function mapping the trainable parameter θ to $(0,1)$; and θ is optimized during training to dynamically adjust the mask's filtering behavio;. $\boldsymbol{F}_{spa}^{(l)}$ denotes the lth layer spatial encoded feature; $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ represent 2-D FFT and IFFT, respectively; $|\cdot|$ extracts Fourier magnitude (intensity information); $\angle\cdot$ extracts Fourier phase (structural information); j is the imaginary unit for complex phase construction; $\mathrm{Re}(\cdot)$ takes the real part of a complex number to obtain real-valued features; $\boldsymbol{F}_{low}^{(l)}$ and $\boldsymbol{F}_{high}^{(l)}$ are the decomposed low- and high-frequency spatial features of $\boldsymbol{F}_{spa}^{(l)}$; and $\odot$ denotes elementwise multiplication.

1) High-Frequency Guided Edge Correction Module: To address the inherent ambiguity of spectral features at land-cover boundaries, the high-frequency guided edge correction module (HGECM) performs edge correction using the high-frequency components of the spatial features. As detailed in Fig. 5, a Sobel operator is first applied to the high-frequency features to explicitly extract an edge response map. This map is then concatenated channelwise with the spectral features to generate a dynamic correction weight map. This weight map adaptively enhances spectral features consistent with spatial edges while suppressing noisy regions, thereby sharpening the boundaries. The edge-corrected spectral features $\boldsymbol{F}_{hs}^{(l)}$ can be expressed as

$$\boldsymbol{F}_{erf}^{(l)} = Conv_{Sobel}\left(\boldsymbol{F}_{high}^{(l)}\right) \tag{7}$$

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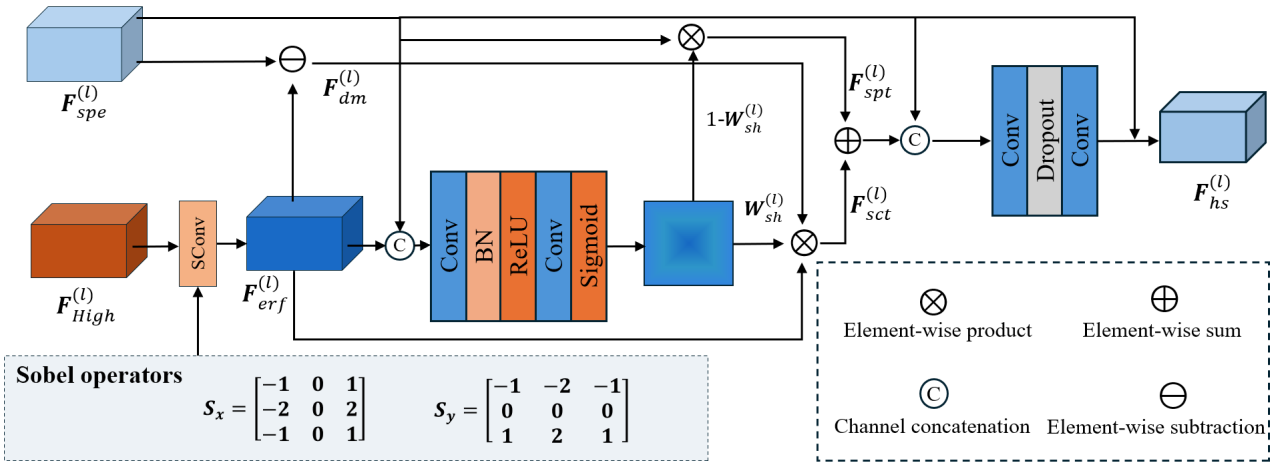


Fig. 5. Detailed architecture of the HGECM.

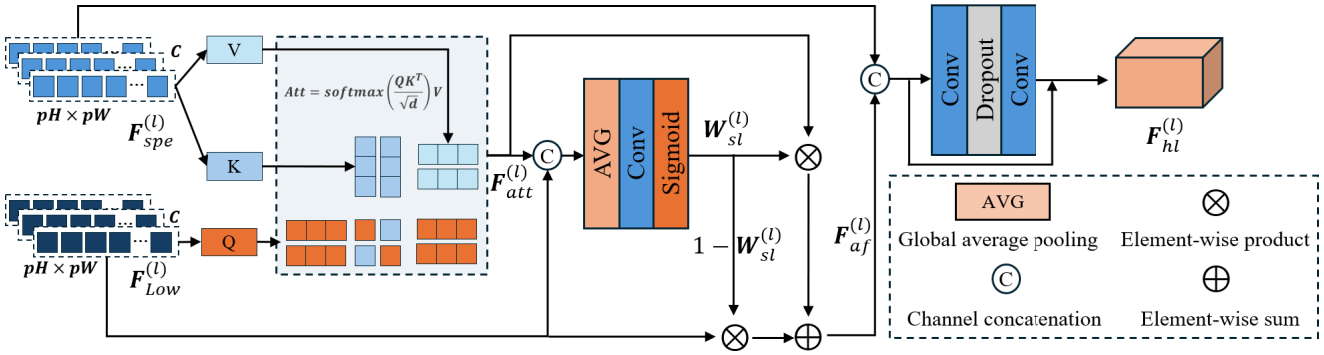


Fig. 6. Detailed architecture of the LGSAM.

$$W_{sh}^{l} = Conv_{1\times1}\left(ReLU\left(BN\left(Conv_{1\times1}\left(Cat\left(F_{erf}^{(l)}, F_{spe}^{(l)}\right)\right)\right)\right)\right) \tag{8}$$

$$F_{spt}^{(l)} = F_{spe}^{(l)} \cdot \left(1 - W_{sh}^{(l)}\right) \tag{9}$$

$$F_{sct}^{(l)} = F_{erf}^{(l)} \cdot W_{sh}^{(l)} \cdot \left(1 + abs\left(F_{spe}^{(l)} - F_{erf}^{(l)}\right)\right) \tag{10}$$

$$F_{hs}^{(l)} = Conv_{3\times3}\left(Dropout\left(Cat\left(F_{spt}^{(l)} + F_{sct}^{(l)}, F_{spe}^{(l)}\right)\right)\right) + F_{spe}^{(l)} \tag{11}$$

where $Conv_{Sobel}(\cdot)$ denotes convolution with the Sobel operator, $abs(\cdot)$ represents the absolute value operation, $Cat(\cdot)$ signifies channelwise concatenation, $Conv_{3\times3}(\cdot)$ refers to a convolution operation, $F_{spt}^{(l)}$ denotes spectral feature proportion, $F_{sct}^{(l)}$ indicates the spatial-guided correction term, $F_{hs}^{(l)}$ corresponds to the output features of the HGECM, $W_{sh}^{(l)}$ stands for the dynamically learned weights, $F_{erf}^{(l)}$ indicates the extracted edge features, and $F_{spe}^{(l)}$ is the spectral features.

2) Low-Frequency Guided Semantic Alignment Module: To achieve region-level semantic consistency, the low-frequency guided semantic alignment module (LGSAM) leverages spatial low-frequency components to guide the alignment of spectral features, effectively reorganizing the spectral feature distribution according to the spatial semantic context. As illustrated in Fig. 6, LGSAM introduces a cross-modal attention mechanism. In particular, the patch sequence from the spatial low-frequency features ($F_{low}^{(l)}$) is used as the query, while the patch sequence from the spectral features ($F_{spe}^{(l)}$) serves as the key and value. By computing attention scores, the module dynamically aggregates the most semantically relevant spectral information for each spatial region. The aligned spectral features are then adaptively fused with the original spatial low-frequency features via a global dynamic weight to generate the final fused features $F_{hl}^{(l)}$

$$W_{sl}^{(l)} = Conv_{1\times1}\left(GlobalAvgPool\left(Cat\left(F_{low}^{(l)}, F_{att}^{(l)}\right)\right)\right) \tag{12}$$

$$F_{af}^{(l)} = W_{sl}^{(l)} \cdot F_{att}^{(l)} + F_{low}^{(l)} \cdot \left(1 - W_{sl}^{(l)}\right) \tag{13}$$

$$F_{hl}^{(l)} = Conv_{3\times3}\left(Dropout\left(Conv_{3\times3}\left(Cat\left(F_{af}^{(l)}, F_{spe}^{(l)}\right)\right)\right)\right) \tag{14}$$

where $W_{sl}^{(l)}$ denotes the dynamic weight, $F_{spe}^{(l)}$ represents the spectral features, $F_{low}^{(l)}$ corresponds to the low-frequency features, $F_{af}^{(l)}$ represents the aligned features, $Cat(\cdot)$ signifies channelwise concatenation, $Conv_{3\times3}(\cdot)$ refers to a convolution operation, $F_{att}^{(l)}$ indicates the attention between the low-frequency features and the original spectral features, and $F_{hl}^{(l)}$ is the output features of the LGSAM.

Ultimately, the fused features produced by FCFM integrate precise boundary details with semantically consistent regional information, providing a solid representational foundation for the auxiliary teacher decoder to generate high-quality predictions.

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D. Pixel Confidence Voting Module

To effectively suppress noise in pseudolabels and alleviate confirmation bias, we construct a PCVM. This module integrates three prediction streams from the spectral teacher, the spatial teacher, and an auxiliary decoder based on FCFM features and generates final high-confidence pseudolabels through a robust voting and filtering process. In this work, we adopt Top-2 entropy to quantify the uncertainty of each pixelwise prediction. Compared with standard entropy, Top-2 entropy focuses on the two most probable classes, making it more sensitive to uncertainty at object boundaries and for hard-to-classify samples. The uncertainty $H_k(x, y)$ of the kth prediction at pixel i is defined as

$$H_k(x,y) = -\left(p_{k,1}(x,y) \times \log\left(p_{k,1}(x,y) + \varphi\right) + p_{k,2}(x,y) \times \log\left(p_{k,2}(x,y) + \varphi\right)\right) \tag{15}$$

where $p_{k,1}(x, y)$ and $p_{k,2}(x, y)$ denote the top two probability values of the softmax output from the kth branch, respectively; φ is a small constant introduced to avoid numerical instability in the logarithm operation; and $H_k(x, y)$ represents the uncertainty of the kth branch at pixel (x, y).

The voting process consists of two steps: initial selection and refinement. In the initial selection step, for each pixel (x, y), the label from the prediction with the lowest uncertainty is chosen as the initial fused label. It is expressed as

$$L^{init}(x,y) = L_{argmin_{k\in\{1,2,3\}}H_k(x,y)}(x,y) \tag{16}$$

where (x, y) denotes the spatial coordinate of a pixel in the image, $L^{init}(x, y)$ represents the initial fused label at pixel (x, y), k is the index of the three prediction branches, $H_k(x, y)$ denotes the uncertainty of the kth branch at pixel (x, y), $L_{argmin_{k\in\{1,2,3\}}H_k(x,y)}(x, y)$ indicates the index of the branch that yields the minimum uncertainty.

In the refinement step, we introduce a confidence threshold τ to filter out unreliable predictions. If the maximum class probability for a pixel across all three prediction branches is lower than τ, that pixel is marked as invalid and ignored during loss computation. This step ensures that only pixels with high consensus and confidence are used as supervisory signals. The process is formulated as

$$L^{mix}(x,y) = \begin{cases} L^{init}(x,y), & \max\left(p_1(x,y), p_2(x,y) p_3(x,y)\right) \geq \tau \\ 255, & \max\left(p_1(x,y), p_2(x,y) p_3(x,y)\right) < \tau \end{cases} \tag{17}$$

where $L^{mix}(x, y)$ denotes the final fused pseudolabel at pixel (x, y), $p_k(x, y)$ represents the maximum class probability of the kth branch at pixel (x, y), and τ denotes the confidence threshold.

E. Multigranularity Consistency Constraints

To fully exploit the potential of unlabeled data and ensure that the student model comprehensively learns from the teachers, we construct a multigranularity consistency loss spanning both the encoder and decoder [73], [74]. This design enforces consistency between the student and teacher models throughout the entire information flow, from shallow features to final predictions.

1) Encoder Feature Consistency: At the encoder level, we design three types of constraints: spectral consistency, spatial consistency, and cross-modal correlation consistency.

Spectral Consistency: Enforced using a combination of cosine similarity and a channel-weighted mean squared error, this loss ensures that the student learns the same material-related features as the spectral teacher, which is expressed as

$$\mathcal{L}_{spe}^{(l)} = \frac{1}{BC}\sum_{b=1}^{B}\sum_{c=1}^{C}\left(1 - \frac{\boldsymbol{F}_{ed}^{s(l)} \cdot \boldsymbol{F}_{ed}^{t(l)}}{\left\|\boldsymbol{F}_{ed}^{s(l)}\right\| \left\|\boldsymbol{F}_{ed}^{t(l)}\right\|}\right) + \frac{1}{BCHW}\sum_{b,c,h,w} \boldsymbol{W}_c^{(l)} \cdot \left(\boldsymbol{F}_{ed}^{s(l)} - \boldsymbol{F}_{ed}^{t(l)}\right)^2 \tag{18}$$

where $\boldsymbol{F}_{ed}^{s(l)}$ denotes the spectral modality features separated from the student features at the lth layer, $\boldsymbol{F}_{ed}^{t(l)}$ refers to the spectral modality features separated from the teacher features at the lth layer, and $\boldsymbol{W}_c^{(l)}$ represents the channel weights obtained by discriminative normalization over all channels at the lth layer.

Spatial Consistency: We employ the Huber loss for local neighborhood continuity and the L1 loss for edge structures to ensure the student preserves geometric details consistent with the spatial teacher, which can be expressed as

$$\mathcal{L}_{spa}^{(l)} = \frac{1}{BHW}\sum_{b,h,w} Huber\left(\boldsymbol{F}_{adh}^{s(l)} - \boldsymbol{F}_{adh}^{t(l)}\right) + \left|M^{s(l)} - M^{t(l)}\right| \tag{19}$$

where $\boldsymbol{F}_{adh}^{s(l)}$ denotes the local spatial neighborhood features of the student at the lth layer; $\boldsymbol{F}_{adh}^{t(l)}$ corresponds to the corresponding local spatial neighborhood features of the teacher at the same layer; and $M^{s(l)}$ and $M^{t(l)}$ represent the gradient magnitudes of the student and teacher features at the lth layer, respectively.

Cross-Modal Correlation Consistency: First, average pooling is applied to the spectral features of the student and teacher models along the height and width dimensions, respectively, to obtain the spectral mean features $\boldsymbol{F}_{edm}^{s(l)}$ and $\boldsymbol{F}_{edm}^{t(l)}$, which aggregate spatial information in the spectral dimension; average pooling is also performed on the spatial features of the student and teacher models along the channel dimension, respectively, to acquire the spatial mean features $\boldsymbol{F}_{adm}^{s(l)}$ and $\boldsymbol{F}_{adm}^{t(l)}$, which aggregate channel information in the spatial dimension. Then, the spectral–spatial correlation features $\boldsymbol{F}_{corr}^{s(l)}$ and $\boldsymbol{F}_{corr}^{t(l)}$ are computed via elementwise multiplication to characterize the cross-modal correlation patterns. Finally, the mse loss is adopted on the correlation matrices between the spectral and spatial features of the student model and the teacher model to force the student model to learn similar intermodal relationships, which is formulated as follows:

$$\boldsymbol{F}_{edm}^{t(l)}, \boldsymbol{F}_{edm}^{s(l)} = \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W} \boldsymbol{F}_{ed}^{t(l)}, \boldsymbol{F}_{ed}^{s(l)} \tag{20}$$

$$\boldsymbol{F}_{adm}^{t(l)}, \boldsymbol{F}_{adm}^{s(l)} = \frac{1}{C}\sum_{c=1}^{C} \boldsymbol{F}_{ad}^{t(l)}, \boldsymbol{F}_{ad}^{s(l)} \tag{21}$$

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$$\boldsymbol{F}_{corr}^{t(l)}, \boldsymbol{F}_{corr}^{s(l)} = \boldsymbol{F}_{spem}^{t(l)} \odot \boldsymbol{F}_{spam}^{t(l)}, \boldsymbol{F}_{spem}^{s(l)} \odot \boldsymbol{F}_{spam}^{s(l)} \tag{22}$$

$$\mathcal{L}_{cross}^{(l)} = \frac{1}{BCHW} \sum_{b,c,h,w} \left(\boldsymbol{F}_{corr}^{s(l)} - \boldsymbol{F}_{corr}^{t(l)}\right)^2 \tag{23}$$

where $\boldsymbol{F}_{corr}^{s(l)}$ and $\boldsymbol{F}_{corr}^{t(l)}$ are the spectral–spatial correlation features of the student and teacher at the lth layer, respectively; $\boldsymbol{F}_{ed}^{s(l)}$ denotes the spectral modality features separated from the student features at the lth layer; $\boldsymbol{F}_{ed}^{t(l)}$ refers to the spectral modality features separated from the teacher features at the lth layer; $\boldsymbol{F}_{edm}^{s(l)}$ and $\boldsymbol{F}_{edm}^{t(l)}$ denote the spectral mean features of the student and teacher features at the lth layer, respectively; and $\boldsymbol{F}_{adm}^{s(l)}$ and $\boldsymbol{F}_{adm}^{t(l)}$ denote the spatial mean features of the student and teacher features at the lth layer, respectively.

The total encoder loss can be expressed as

$$\mathcal{L}_{ecc} = \frac{1}{L} \sum_{l=0}^{L-1} mask\,(l) \cdot \left(w_{spe}\,(l)\,\mathcal{L}_{spe}^{(l)} + w_{spa}\,(l)\,\mathcal{L}_{spa}^{(l)} + w_{cross}\,(l)\,\mathcal{L}_{cross}^{(l)}\right) \tag{24}$$

where $w_{spe}(l), w_{spa}(l)$, and $w_{cross}(l)$ denote the dynamic weight assigned to each layer, $\mathcal{L}_{spe}^{(l)}$ represents the spectral modality consistency loss at the lth layer, $\mathcal{L}_{spa}^{(l)}$ corresponds to the spatial modality consistency loss between the student and teacher at the lth layer, $\mathcal{L}_{cross}^{(l)}$ indicates the spectral–spatial correlation loss at the lth layer, and $\mathcal{L}_{ecc}$ is the total encoder loss at layer L.

2) Decoder Structure Consistency: At the decoder stage, in addition to the standard cross-entropy loss, we introduce semantic and structural consistency losses.

Semantic Feature Consistency: The Kullback–Leibler (KL) divergence is used to constrain the consistency between the channelwise distributions of the student and teacher decoder outputs, ensuring high-level semantic alignment. This loss is formulated as

$$\mathcal{L}_{sem}^{(l)} = P\alpha^l \cdot KL\left(P^{s(l)} \| P^{t(l)}\right) + \left(1-\alpha^l\right) \cdot \left(1 - Mean_{B,H,W}\left(cosine\left(\boldsymbol{F}^{s(l)}, \boldsymbol{F}^{t(l)}\right)\right)\right) \tag{25}$$

where $P^{s(l)}$ and $P^{t(l)}$ denote the channelwise distributions of the student and teacher at the lth layer, respectively; $\boldsymbol{F}^{s(l)}$ and $\boldsymbol{F}^{t(l)}$ represent the reshaped aligned features of the student and teacher at the same layer; $Mean_{B,H,W}(\cdot)$ indicates averaging over the batch and spatial dimensions; and $KL(\cdot)$ is the KL divergence.

Spatial Structure Consistency: This term combines boundary gradient constraints (cosine similarity and L1 loss) with local region consistency (mean squared error loss) to ensure the geometric structure of the student's prediction maps remains consistent with the teacher's. It can be expressed as

$$\mathcal{L}_{spa}^{(l)} = \frac{1}{BHW} \sum_{b=1}^{B} \sum_{h=1}^{H} \sum_{w=1}^{W} \left(\left|\|\nabla \boldsymbol{F}_{align}^{s(l)}\|_2 - \|\nabla \boldsymbol{F}_{align}^{t(l)}\|_2\right| - \cos\theta_{b,h,w}\right) + \beta \cdot MSE\left(\boldsymbol{F}_{align}^{s(l)}, \boldsymbol{F}_{align}^{t(l)}\right) \tag{26}$$

where $\cos\theta_{b,h,w}$ denotes the angle between the gradient vectors of the student and the teacher; $\|\nabla \boldsymbol{F}_{align}^{s(l)}\|_2$ and $\|\nabla \boldsymbol{F}_{align}^{t(l)}\|_2$ represent the gradient magnitudes of the aligned student and teacher features, respectively; $\boldsymbol{F}_{align}^{s(l)}$ and $\boldsymbol{F}_{align}^{t(l)}$ correspond to the aligned student and teacher features at the lth layer; $mse(\cdot)$ refers to the mean squared error; and β is the weight of the local region loss.

Hierarchical Semantic Gradient: This loss enforces consistency in the feature differences between hierarchically upsampled layers, promoting smooth and coherent feature evolution through the decoder. It is formulated as

$$\mathcal{L}_{grad}^{(l)} = \frac{1}{L-1} \sum_{i=0}^{L-2} \frac{1}{BCHW} \sum_{b,c,h,w} \left| \left(\boldsymbol{F}^{s(l+1)} - I\left(\boldsymbol{F}^{s(l)}\right)\right) - \left(\boldsymbol{F}^{t(l+1)} - I\left(\boldsymbol{F}^{t(l)}\right)\right) \right| \tag{27}$$

where $L_{spa}^{(l)}$ and $L_{sem}^{(l)}$ denote the student decoder features at the lth and $(l+1)$th layers, respectively; and $I(\cdot)$ indicates the upsampling operation.

The total decoder loss can be expressed as

$$\mathcal{L}_{dcc} = \frac{1}{T}\frac{1}{L} \sum_{l=0}^{L-1} \sum_{t=1}^{T} \left(w_{spa}^{(l)} \times \mathcal{L}_{spa}^{t(l)} + w_{sem}^{(l)} \times \mathcal{L}_{sem}^{t(l)}\right) + \mathcal{L}_{grad} \tag{28}$$

where $\mathcal{L}_{spa}^{t(l)}$ and $\mathcal{L}_{sem}^{t(l)}$ denote the spatial and semantic losses at the lth layer under the tth perturbation, respectively; $\mathcal{L}_{grad}$ indicates the hierarchical semantic gradient loss; $w_{spa}(l)$ and $w_{sem}(l)$ are layerwise dynamic weights; and T signifies the number of small noise perturbations applied to the aligned student features.

IV. Experiments Results

This section systematically evaluates the performance of the proposed DDCSF on two representative public benchmark datasets. First, the datasets, implementation details, and evaluation metrics are presented. Then, DDCSF is compared quantitatively and qualitatively with state-of-the-art semisupervised semantic segmentation methods. Furthermore, a series of ablation studies analyze the effectiveness of each designed component. Finally, model complexity and convergence behavior are discussed.

A. Datasets

Two representative public datasets, ISPRS Potsdam [75] and GID [76], are used for experimental evaluation.

ISPRS Potsdam Dataset: This dataset comprises 38 high-resolution aerial images of Potsdam, Germany, with a ground sampling distance (GSD) of 5 cm. Each image has a size of 6000 × 6000 pixels and includes five spectral bands: red (R), green (G), blue (B), NIR, and a digital surface model (DSM). The normalized difference vegetation index (NDVI) is also estimated and used as an additional channel. The dataset covers six categories: impermeable surfaces, buildings, low vegetation, trees, cars, and clutter/background. For training, validation, and testing, all images were cropped into nonoverlapping patches of 512 × 512 pixels.

GID Dataset: A benchmark for land use classification, comprising 150 high-resolution satellite images from over 60 cities

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in China. The spatial resolution is 1 m for the panchromatic band and 4 m for the multispectral bands (NIR, R, G, and B). Each image is 6800 × 7200 pixels. The dataset covers seven land cover categories: buildings, farmland, forests, grasslands, water bodies, and bare land with clutter. All images were cropped into 512 × 512 patches, totaling 43 200 image patches.

B. Experimental Details

All experiments were conducted using the PyTorch framework on a workstation equipped with an NVIDIA RTX 3090 GPU (24 GB VRAM). For the semisupervised setting, we trained the model using three different proportions of labeled data: 1%, 5%, and 10%. We adopted a weak-to-strong augmentation strategy for consistency learning. For unlabeled data, weak augmentation included only random flipping and cropping, while strong augmentation employed more aggressive strategies such as CutMix to encourage the model to learn more robust feature representations.

We used the stochastic gradient descent (SGD) optimizer with momentum for training. The initial learning rate was set to 0.01, momentum to 0.9, and weight decay to 0.0001. A polynomial learning rate decay schedule was employed, and the model was trained for 300 epochs.

For the uncertainty calculation in the PCVM module, a small constant $\varphi = 10^{-8}$ is added to the probability values to avoid numerical instability caused by logarithmic operations and prevent invalid log(0) calculations. The value of φ is fixed to 10^{-8} in all experiments of this article.

Three standard metrics were used for performance evaluation: mean value of joint intersection (mIoU) [77], mean score of $F1$ (m$F1$) [78], and Kappa coefficient [79].

C. Comparison With Other Methods

To evaluate the performance of DDCSF, we conduct rigorous quantitative comparisons with a variety of state-of-the-art semisupervised semantic segmentation methods. These include approaches originally designed for natural images, such as mean teacher [41], CutMix [69], CCT [70], FixMatch [44], and CPS [45] as well as recent methods specifically developed for remote sensing scenarios, including LSST [71], CCVC [72], UniMatch [43], DWL [47], and MUCA [46].

1) Quantitative Analysis: Tables I and II report the detailed quantitative comparison results of different methods on the two datasets under varying labeling ratios. Bold numbers indicate the best performance, while underlined numbers denote the second-best results. The following observations can be made from these tables.

On the ISPRS Potsdam dataset, DDCSF achieves the best performance across all labeling ratios. Notably, under the extremely sparse setting with only 1% labeled data, DDCSF attains an mIoU of 67.95%, an m$F1$ of 78.93%, and a Kappa coefficient of 0.8130, outperforming the second-best method, MUCA, by 1.78%, 2.65%, and 2.33%, respectively. It is worth noting that DDCSF exhibits particularly strong performance on the tree and low vegetation classes, which are spectrally similar and texturally alike. For instance, with 10% labeled data, DDCSF achieves IoU scores of 67.19% and 68.20% for trees and low vegetation, respectively. This demonstrates the effectiveness of the proposed spectral–spatial decoupling and collaborative learning mechanism: the spectral teacher accurately captures subtle material differences among vegetation types, while the spatial teacher preserves the structural integrity of vegetation canopies. Their collaboration ultimately enables precise class discrimination.

On the GID dataset, DDCSF also achieves the best results. With only 1% labeled data, DDCSF reaches an mIoU of 63.13%, an m$F1$ of 76.29%, and a Kappa coefficient of 71.12%, again outperforming all competing methods. In particular, for the water class, DDCSF achieves an IoU of 78.63%, exceeding the second-best method by 1.31%. This improvement can be attributed to the adaptive enhancement of spectral features by ASSIFM and the deep spectral–spatial alignment enabled by the FCFM module, which allow the model to effectively exploit the distinctive spectral signatures of water bodies. Even in complex backgrounds, DDCSF can accurately localize water boundaries, thereby generating semantically cleaner pseudolabels.

2) Qualitative Analysis: Figs. 7 and 8 present qualitative visual comparisons of segmentation results on the ISPRS Potsdam and GID datasets, respectively. As shown in Fig. 7, many methods (e.g., CPS and UniMatch) suffer from noticeable confusion and misclassification along the boundaries between low vegetation and trees, particularly in regions with dense buildings and interwoven vegetation. On the contrary, the segmentation maps produced by DDCSF [Fig. 7(h)] exhibit clear and smooth boundaries between vegetation categories, as well as more regular building outlines with fewer internal holes. This improvement primarily stems from the edge correction of spectral features guided by the high-frequency components in FCFM, combined with the effective filtering of uncertain boundary predictions by PCVM. Together, these components lead to geometrically more accurate and semantically more reliable pseudolabels.

As illustrated in Fig. 8, for large-area segmentation of farmland, forest, and water bodies, DDCSF demonstrates superior region integrity and boundary clarity. For example, when segmenting water bodies, other methods may produce fragmented or incomplete masks, whereas DDCSF generates continuous and coherent predictions [Fig. 8(h)]. Moreover, the boundaries between water bodies and farmland are noticeably sharper. These results further verify that our proposed dual-teacher decoupling and collaborative framework is capable of generating high-quality pseudolabels that are both semantically consistent and structurally complete, thereby effectively guiding the student model and alleviating small-region errors and boundary ambiguity.

D. Ablation Studies

To investigate the contribution of each core component in DDCSF, we conducted a series of detailed ablation experiments on the ISPRS Potsdam dataset with a 5% labeled-data setting.

1) Contribution of Different Module Combinations to DDCSF: We performed experiments using different combinations of our proposed modules. The results are reported in

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TABLE I
SEGMENTATION RESULTS OF DIFFERENT METHODS ON THE ISPRS POTSDAM DATASET

Ratio	Models	IoU						mIoU/mF1/Kappa
		Imp.surp	Building	Low.Veg	Tree	Car	Clutter	
1%	Mean teacher [41]	70.37	80.71	58.60	60.33	67.21	54.00	65.20 / 75.36 / 0.7787
	CutMix [69]	69.70	78.40	59.89	56.61	66.91	51.89	63.90 / 73.66 / 0.7690
	CCT [70]	70.44	78.42	61.60	58.21	67.41	53.51	64.93 / 75.00 / 0.7753
	FixMatch [44]	71.00	79.90	61.36	59.53	68.50	54.71	65.83 / 75.83 / 0.7822
	CPS [45]	69.33	81.01	59.52	60.72	66.93	**56.80**	65.72 / 75.90 / 0.7857
	LSST [71]	69.87	81.22	58.73	58.60	67.53	54.22	65.03 / 75.14 / 0.7761
	CCVC [72]	70.21	80.31	59.37	62.17	66.23	53.99	65.38 / 75.53 / 0.7813
	UniMatch [43]	67.52	85.02	59.73	**63.44**	65.61	54.70	66.00 / 76.31 / 0.7869
	DWL [47]	67.61	80.73	61.55	59.32	66.09	54.00	64.88 / 74.88 / 0.7739
	MUCA [46]	70.32	81.31	62.07	60.67	65.89	56.73	66.17 / 76.28 / 0.7897
	DDCSF(Ours)	**71.27**	**85.32**	**63.79**	63.33	**67.90**	56.11	**67.95 / 78.93 / 0.8130**
5%	Mean teacher [41]	72.58	82.33	60.45	62.56	69.01	56.13	67.18 / 77.55 / 0.8036
	CutMix [69]	71.86	80.71	62.91	58.78	68.70	54.63	66.27 / 76.39 / 0.7910
	CCT [70]	72.71	80.87	63.33	60.39	69.09	55.78	67.03 / 77.85 / 0.8027
	FixMatch [44]	72.55	81.12	63.87	61.45	70.32	56.47	67.63 / 78.41 / 0.8092
	CPS [45]	71.21	83.17	62.01	62.14	68.77	**58.47**	67.63 / 78.47 / 0.8120
	LSST [71]	72.34	83.03	60.69	60.32	69.78	56.42	67.10 / 77.96 / 0.8031
	CCVC [72]	72.35	82.68	61.32	63.78	68.75	55.97	67.48 / 78.33 / 0.8060
	UniMatch [43]	70.12	**87.22**	61.93	61.96	69.21	57.12	67.93 / 78.82 / 0.8111
	DWL [47]	70.33	82.55	63.80	61.67	68.33	55.22	66.98 / 77.10 / 0.7999
	MUCA [46]	72.40	83.09	63.43	62.66	68.93	58.07	68.10 / 79.19 / 0.8103
	DDCSF(Ours)	**73.34**	87.05	**65.39**	**64.02**	**70.33**	58.34	**69.75 / 80.74 / 0.8318**
10%	Mean teacher [41]	75.23	83.71	63.21	65.22	**73.72**	56.93	69.67 / 80.60 / 0.8322
	CutMix [69]	72.90	84.42	65.73	60.69	70.61	58.27	68.77 / 79.68 / 0.8209
	CCT [70]	73.88	85.53	66.10	63.27	71.51	57.22	69.59 / 80.51 / 0.8312
	FixMatch [44]	75.80	84.35	66.41	64.71	71.73	58.72	70.29 / 81.21 / 0.8383
	CPS [45]	74.19	86.03	65.43	65.11	69.93	60.21	70.15 / 81.04 / 0.8361
	LSST [71]	75.30	86.37	63.44	62.19	70.91	59.55	69.63 / 80.55 / 0.8319
	CCVC [72]	75.23	85.33	64.23	66.82	71.43	58.30	70.22 / 81.17 / 0.8366
	UniMatch [43]	73.60	87.81	64.70	63.77	70.78	60.83	70.25 / 81.17 / 0.8372
	DWL [47]	72.01	85.22	66.74	64.32	70.54	58.31	69.52 / 80.44 / 0.8303
	MUCA [46]	74.37	87.20	66.55	65.80	71.90	**61.22**	71.17 / 82.05 / 0.8460
	DDCSF(Ours)	**76.21**	**90.11**	**68.20**	**67.19**	73.60	60.01	**72.55 / 83.47 / 0.8601**
100%	FullySup	78.91	92.12	70.56	70.01	74.92	63.16	74.95 / 85.90 / 0.8821

Table III and visualized in Fig. 9. A baseline model consisting only of the dual-teacher architecture achieves an mIoU of 67.52%. When either FCFM or ASSIFM is individually added, the mIoU increases to 68.61% (+1.09%) and 68.33% (+0.81%), respectively, demonstrating the effectiveness of each module in isolation. When both modules are integrated, the full DDCSF model attains an mIoU of 69.75%, a 2.23% improvement over the baseline. As illustrated in Fig. 9, the baseline model produces a large number of erroneous predictions along building boundaries and within building interiors. With the gradual incorporation of our modules, these error regions are significantly reduced. The final DDCSF model markedly alleviates internal holes and boundary blurring for large-scale objects, indicating that FCFM and ASSIFM complement and reinforce each other to jointly improve segmentation performance.

2) Impact of FCFM on the Performance of DDCSF: This section investigates the effect of the FCFM on model performance. First, we replaced FCFM in DDCSF with several alternative fusion modules and trained the models under identical conditions. The results are reported in Table IV. Replacing FCFM with any of the alternative modules leads to performance degradation. DDCSF equipped with our FCFM achieves the best performance, with an mIoU of 69.75%, outperforming the second-best method SSMA by 1.42%. These results clearly demonstrate the superiority of the FCFM module.

Next, we conducted an ablation study on the internal components of FCFM, with results summarized in Table V. Removing both HGECM and LGSAM causes the mIoU to drop to 67.29%. Although using either module alone yields some improvement, the gains are limited. This is because edge

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TABLE II
SEGMENTATION RESULTS OF DIFFERENT METHODS ON THE GID DATASET

Ratio	Model	IoU						mIoU/mFI/Kappa
		Built-up	Water	Forest	Meadow	Farmland	Others	
1%	Mean teacher [41]	55.31	75.61	59.32	42.87	70.33	51.62	59.18 / 72.06 / 0.6693
	CutMix [69]	54.93	75.80	58.73	42.36	70.20	51.00	58.84 / 71.93 / 0.6667
	CCT [70]	55.42	75.00	59.52	43.61	71.02	52.93	59.58 / 72.49 / 0.6731
	FixMatch [44]	54.73	76.22	63.17	46.42	72.63	53.77	61.16 / 74.22 / 0.6914
	CPS [45]	55.79	75.21	59.77	43.33	71.12	53.39	59.77 / 72.74 / 0.6758
	LSST [71]	56.33	75.37	60.29	43.53	71.52	53.47	60.09 / 73.07 / 0.6792
	CCVC [72]	57.33	75.18	61.33	43.87	71.62	53.58	60.40 / 73.49 / 0.6839
	UniMatch [43]	**57.33**	75.72	61.47	44.73	71.33	53.62	60.70 / 73.81 / 0.7011
	DWL [47]	55.56	76.29	63.64	47.13	73.14	54.11	61.65 / 75.23 / 0.6966
	MUCA [46]	55.76	77.32	63.42	46.72	**74.92**	54.27	62.07 / 75.17 / 0.7005
	DDCSF(Ours)	57.24	**78.63**	**64.78**	**47.62**	74.63	**55.90**	**63.13 / 76.29 / 0.7112**
5%	Mean teacher [41]	56.26	77.32	63.20	46.55	73.10	54.44	61.81 / 75.21 / 0.7045
	CutMix [69]	55.78	76.51	62.32	45.70	72.26	53.22	60.97 / 74.10 / 0.6923
	CCT [70]	57.37	76.22	63.64	47.56	73.33	55.32	62.24 / 75.36 / 0.7025
	FixMatch [44]	57.91	77.43	64.91	49.21	73.72	57.01	63.37 / 76.49 / 0.7201
	CPS [45]	57.45	76.45	63.23	47.56	73.00	55.73	62.24 / 75.33 / 0.7021
	LSST [71]	56.51	77.03	63.76	47.99	73.26	56.11	62.44 / 75.54 / 0.7044
	CCVC [72]	57.11	77.44	64.57	48.33	73.73	56.51	62.95 / 76.05 / 0.7093
	UniMatch [43]	57.41	77.92	65.00	48.61	74.21	56.98	63.36 / 76.49 / 0.7132
	DWL [47]	56.79	78.56	65.33	48.92	75.67	57.66	63.82 / 77.04 / 0.7180
	MUCA [46]	56.33	79.71	64.41	**50.31**	74.73	**57.90**	63.90 / 77.15 / 0.7191
	DDCSF(Ours)	**59.41**	**80.96**	**66.91**	49.53	**76.80**	57.89	**65.25 / 78.44 / 0.7323**
10%	Mean teacher [41]	56.73	77.95	63.88	47.75	73.76	55.53	62.60 / 75.75 / 0.7115
	CutMix [69]	56.44	77.47	63.41	46.23	73.48	54.45	61.91 / 75.20 / 0.7035
	CCT [70]	57.43	77.55	65.85	48.99	73.82	55.62	63.21 / 76.52 / 0.7129
	FixMatch [44]	58.00	77.52	65.24	50.13	73.93	57.55	63.73 / 77.04 / 0.7177
	CPS [45]	57.53	77.10	65.15	48.43	73.56	55.81	62.93 / 76.55 / 0.7089
	LSST [71]	57.53	77.42	66.44	48.51	74.17	56.83	63.48 / 76.71 / 0.7152
	CCVC [72]	58.26	78.63	66.60	48.79	74.55	57.46	64.05 / 77.32 / 0.7215
	UniMatch [43]	59.54	79.83	66.51	50.33	75.90	57.73	64.97 / 78.20 / 0.7307
	DWL [47]	59.82	80.72	66.71	50.43	76.52	58.77	65.50 / 78.73 / 0.7369
	MUCA [46]	59.62	81.90	67.33	50.95	77.81	59.53	66.19 / 79.36 / 0.7430
	DDCSF(Ours)	**61.67**	**83.43**	**68.83**	**52.81**	**78.76**	**60.58**	**67.68 / 80.85 / 0.7572**
100%	FullySup	64.27	85.79	72.83	56.20	82.93	63.52	70.92 / 84.11 / 0.7916

TABLE III
ABLATION RESULTS OF THE CORE MODULES IN DDCSF

FCFM	ASSIFM	mIoU/mFI/Kappa
		67.52 / 78.92 / 0.8211
✓		68.33 / 79.72 / 0.8169
	✓	68.61 / 79.65 / 0.8204
✓	✓	**69.75 / 80.74 / 0.8318**

TABLE IV
IMPACT OF DIFFERENT FEATURE FUSION MODULES ON DDCSF

Methods	mIoU/mF1/Kappa
SAFFM [80]	67.52 / 78.42 / 0.8083
SSR-NET [81]	67.83 / 78.66 / 0.8106
DFM [59]	68.08 / 79.00 / 0.8141
SSMA [58]	68.33 / 79.15 / 0.8171
SSFBF [57]	68.17 / 79.16 / 0.8155
FCFM(Ours)	**69.75 / 80.74 / 0.8318**

correction alone lacks region-level semantic guidance, while semantic alignment alone may overlook fine boundary details. Only when both modules operate collaboratively does the model achieve the best performance (mIoU of 69.75%). These findings indicate that by leveraging high- and low-frequency components for boundary refinement and semantic alignment, respectively, FCFM enables a complementary integration of structural and semantic information, thereby producing optimal fused features.

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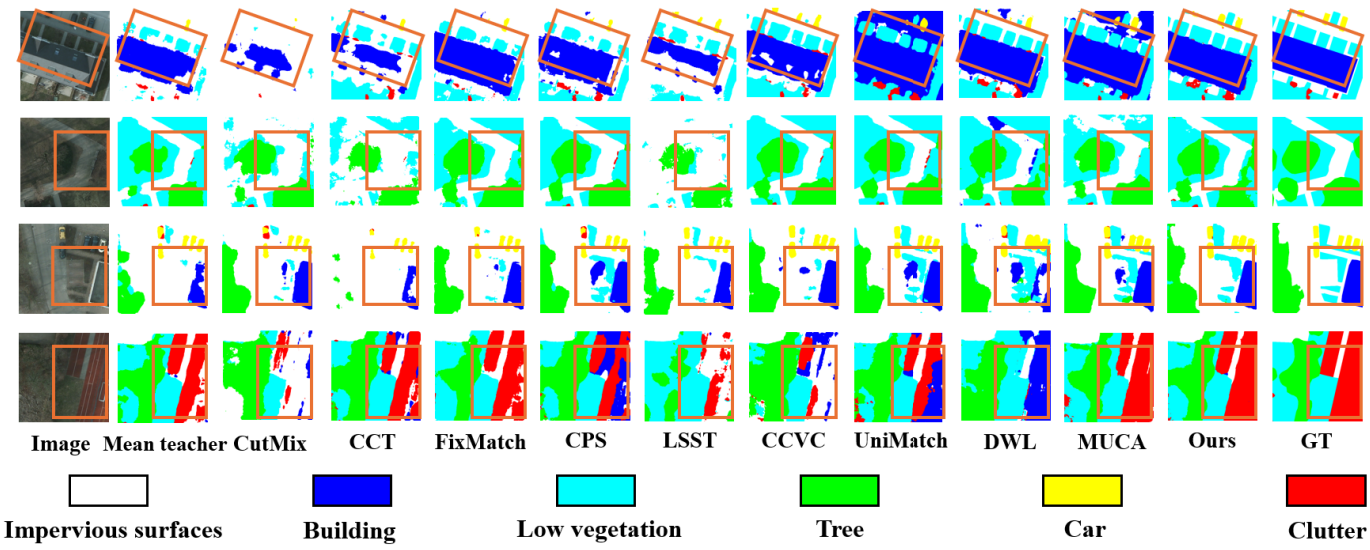


Fig. 7. Example segmentation results of different methods on the ISPRS Potsdam dataset.

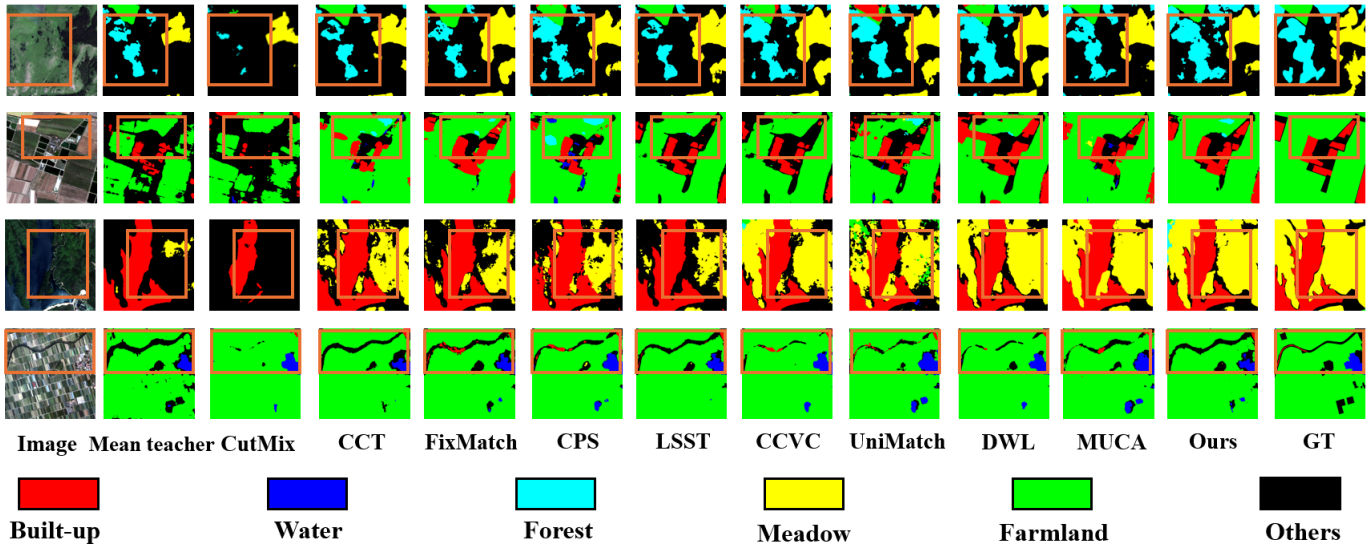


Fig. 8. Example segmentation results of different methods on the GID dataset.

TABLE V
RESULTS OF DIFFERENT COMBINATIONS OF LGSAM AND HGECM IN FCFM

LGSAM	HGECM	mIoU/mF1/Kappa
		67.29 / 78.31 / 0.8067
✓		68.63 / 79.65 / 0.8200
	✓	68.52 / 79.52 / 0.8192
✓	✓	**69.75 / 80.74 / 0.8318**

TABLE VI
RESULTS OF LGSAM WITH DIFFERENT PATCH SIZES

Patch	mIoU/mF1/Kappa
1×1	68.96 / 79.93 / 0.8233
2×2	69.33 / 80.62 / 0.8291
4×4	**69.75 / 80.74 / 0.8318**
6×6	69.44 / 80.50 / 0.8298
8×8	68.57 / 79.61 / 0.8195

3) Parameter Sensitivity Analysis of LGSAM: In LGSAM, the patch size is a key hyperparameter that affects performance, as it determines the granularity of region-level alignment. Table VI reports the impact of different patch sizes. When the patch size is too small (e.g., 1×1), the model fails to capture sufficient regional context. Conversely, when the patch size is too large (e.g., 8×8), local structural information can be disrupted. A patch size of 4×4 achieves the optimal balance, yielding the best performance across all metrics (mIoU: 69.75%). This setting was therefore adopted in all experiments.

4) Validation of the Multisource Pseudolabel Generation Strategy: To verify the effectiveness of our multisource

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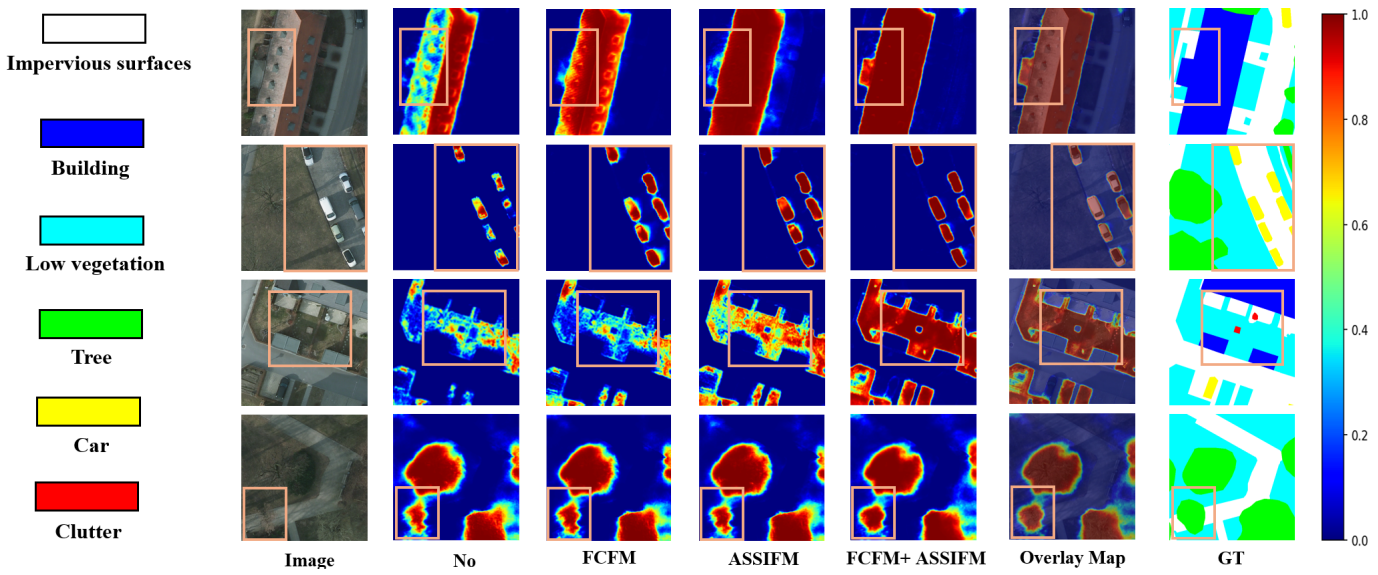


Fig. 9. Heatmap visualization of segmentation results with different module combinations.

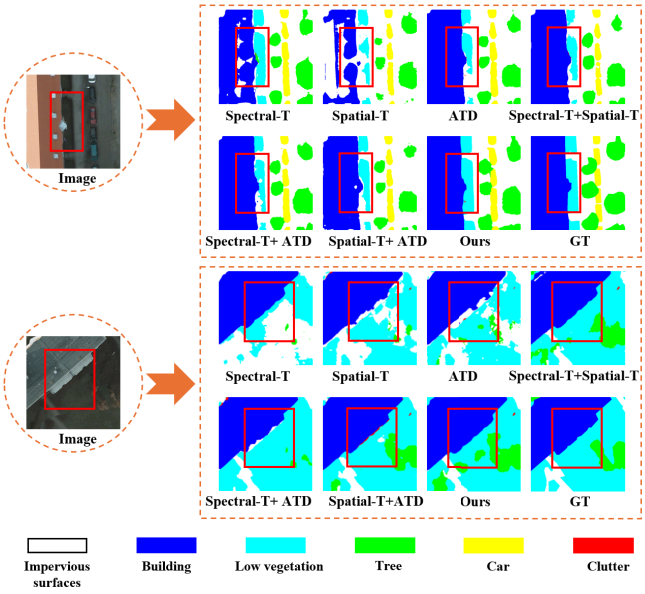


Fig. 10. Mixed pseudolabels generated by different combinations.

TABLE VII

RESULTS OF DIFFERENT COMBINATIONS OF THE DUAL TEACHERS AND THE AUXILIARY DECODER

Spectral-T	Spatial-T	ATD	mIoU/mF1/Kappa
✓			66.91 / 78.31 / 0.8033
	✓		67.33 / 78.68 / 0.8055
		✓	67.45 / 78.85 / 0.8209
✓	✓		68.93 / 80.35 / 0.8236
✓		✓	68.25 / 79.67 / 0.8160
	✓	✓	68.59 / 79.56 / 0.8251
✓	✓	✓	**69.75 / 80.74 / 0.8318**

pseudolabel generation strategy (dual teachers + auxiliary decoder), we conducted a series of experiments with results reported in Table VII and visualized in Fig. 10. When pseudolabels are generated from any single source alone, the student model's performance is limited. As more prediction sources are incorporated, both the pseudolabel quality and the final segmentation accuracy steadily improve. When all three branches collaborate through the PCVM, the student model achieves an mIoU of 69.75%, significantly outperforming any dual-source combination. This demonstrates that our multisource voting mechanism effectively suppresses the inherent bias of individual models by cross-validating and aggregating complementary strengths, thereby substantially improving pseudolabel quality and reliability.

Fig. 10 visually confirms these findings. The pseudolabels from the spectral teacher show superior material discrimination, while those from the spatial teacher provide clearer geometric contours. The auxiliary decoder's predictions achieve the best object completeness but have slightly blurred boundaries. The final pseudolabels produced by PCVM [Fig. 10(d)] successfully integrate these strengths, preserving both structural completeness and sharp, well-defined boundaries.

5) Quantitative Analysis of Pseudolabel Accuracy at Different Training Stages: To quantitatively verify that the proposed method improves pseudolabel quality, we conduct comparative experiments from two dimensions: the proportion of valid pseudopixels and the visual performance of pseudolabels. Four mainstream semisupervised approaches, namely FixMatch, CPS, UniMatch, and MUCA, are adopted as baselines. We continuously record the proportion of valid pseudopixels filtered by each method over 0–100 training epochs, and present the pseudolabel outputs generated by different models at the 100th training epoch.

As illustrated in Fig. 11, the proportion of valid pseudopixels of all methods rises steadily as training proceeds. At the early training stage, the model suffers from underfitting and high prediction uncertainty, resulting in low-quality pixels and a valid pseudopixel ratio of only 57%–62%. As the model gradually converges, pseudolabel quality is continuously improved, and more valid pixels become available for

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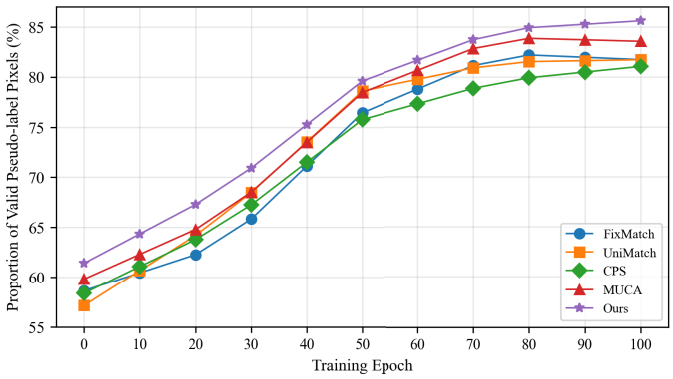


Fig. 11. Comparison of the ratio of valid pseudolabel pixels between the proposed DDCSF and baseline methods across training epochs.

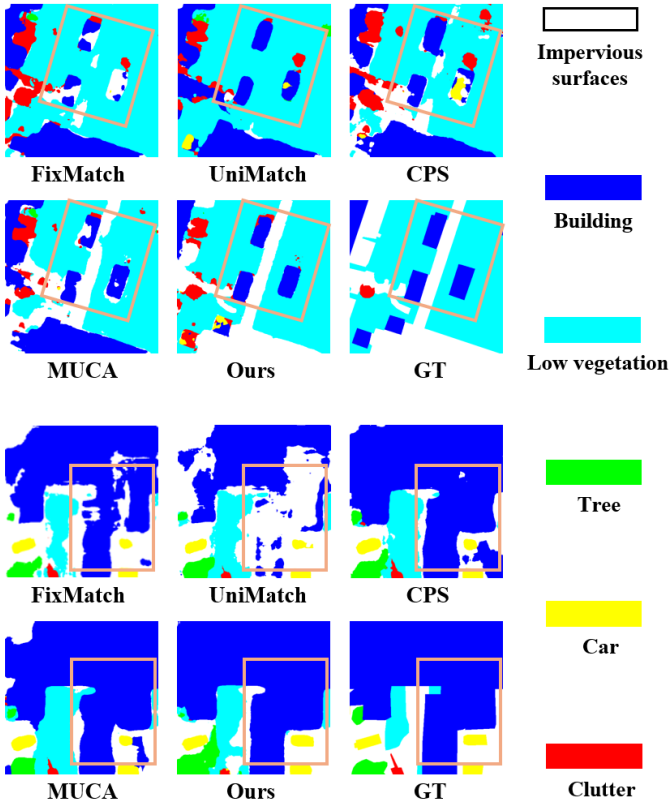


Fig. 12. Visual comparison of pseudolabels generated by the proposed DDCSF and baseline methods at training epoch 100.

supervised training. Horizontal comparison of all curves shows that the curve of our DDCSF lies above all baselines throughout the training process. In the early stage (Epoch 0–40), DDCSF can already filter out more high-confidence pixels. The performance gap keeps widening in the middle and late training phases. At Epoch 100, the proportion of valid pseudopixels of DDCSF reaches approximately 85.75%, which is remarkably higher than MUCA, UniMatch, CPS, and FixMatch.

To quantitatively and intuitively compare the quality of pseudolabels generated by different methods, apart from calculating the proportion of valid pixels at each training stage, we further visualize the pseudolabels produced by models trained

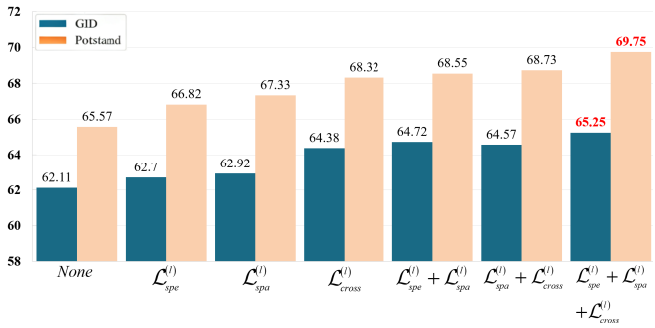


Fig. 13. Results of different loss combinations for encoder-side constraints.

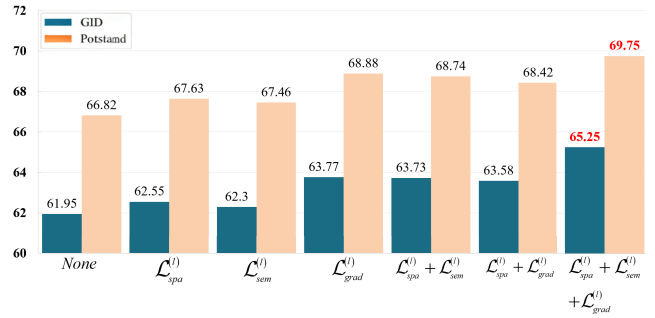


Fig. 14. Results of different loss combinations for decoder-side constraints.

for 100 epochs, as shown in Fig. 12. Typical regions with confusing categories and complex boundaries, such as buildings and vegetation, are selected for demonstration. Baseline methods including FixMatch, UniMatch, CPS, and MUCA generate predictions relying merely on single-modal frameworks or simple dual-branch architectures. Obvious misclassifications, holes, and fragmented noise exist at vegetation boundaries with similar spectra and inside building areas. On the contrary, pseudolabels output by the proposed DDCSF feature complete and clear ground object contours with sharp segmentation boundaries between vegetation and buildings, which are highly consistent with ground-truth annotations. The visualization results directly verify that PCVM can effectively alleviate pseudolabel noise caused by modal collapse and generate structurally intact and semantically accurate supervision signals under the same confidence filtering criteria.

6) Impact of MGCCs on DDCSF Performance: To verify the effectiveness of the multigranularity consistency loss, we conducted ablation experiments on the encoder and decoder constraints. The results are shown in Figs. 13 and 14. As can be observed, removing any of the constraints, whether spectral, spatial, or cross-modal consistency at the encoder, or structural, semantic, or hierarchical gradient consistency at the decoder, leads to a degradation in performance. Only when all constraints operate jointly does the model achieve the most comprehensive alignment in both the feature and output spaces, thereby maximizing the use of supervisory signals from unlabeled data. These results demonstrate that the proposed multigranularity consistency loss is integral to the model's performance.

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TABLE VIII
TIME PER ITERATION

Method	Training time(s)	mIoU/mF1/Kappa
Mean teacher	69	67.18 / 77.55 / 0.8036
CutMix	78	66.27 / 76.39 / 0.7910
CCT	74	67.03 / 77.85 / 0.8027
FixMatch	**62**	67.63 / 78.41 / 0.8092
CPS	70	67.63 / 78.47 / 0.8120
LSST	83	67.10 / 77.96 / 0.8031
CCVC	77	67.48 / 78.33 / 0.8060
UniMatch	67	67.93 / 78.82 / 0.8111
DWL	86	66.98 / 77.10 / 0.7999
MUCA	83	68.10 / 79.19 / 0.8103
DDCSF(Ours)	96	**69.75 / 80.74 / 0.8318**

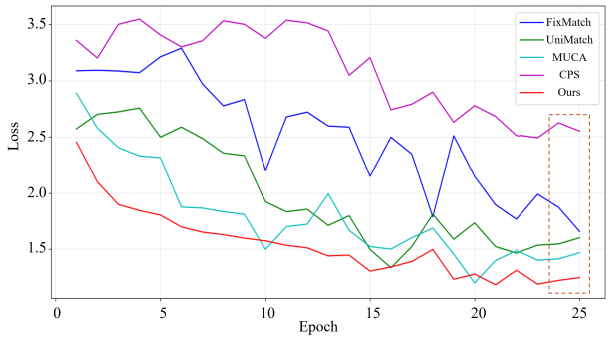


Fig. 15. Convergence of the first 25 training epochs is presented.

E. Complexity Analysis

Finally, we evaluated the computational overhead of DDCSF.

The computational complexity is shown in Table VIII. This section compares the training time of different methods in a single iteration (one epoch). With the dual-teacher architecture and the extra collaborative fusion module, DDCSF has a longer training time (96 s) than other methods. Nevertheless, this increased training cost brings considerable performance improvements. For example, its mIoU is 2.12% higher than that of the fastest FixMatch and 1.82% higher than that of the second-fastest UniMatch.

The convergence is illustrated in Fig. 15. As can be seen from the figure, DDCSF not only presents a higher initial performance baseline at the early convergence stage but also maintains a stable convergence process, and the final performance plateau it reaches is significantly higher than that of all comparative methods. This indicates that DDCSF can exploit unlabeled data more efficiently and stably, and converge rapidly to a more optimal solution, which further validates the superiority of the proposed framework.

V. CONCLUSION

We proposed the DDCSF to address data scarcity in remote sensing image segmentation. The core of DDCSF lies in its specialized spectral and spatial teachers, whose outputs are integrated through a collaborative mechanism. This is achieved using key components: a FCFM for precise boundary and semantic fusion, a PCVM for robust pseudolabel selection, and an ASSIFM to provide the student model with enriched input. Extensive experiments on the ISPRS Potsdam and GID benchmarks validate that DDCSF achieves state-of-the-art performance. Despite its success, limitations include a fixed-ratio frequency decoupling and the computational cost of the dual-teacher architecture. In the future, we will focus on developing more adaptive frequency-domain mechanisms and exploring model lightweighting strategies. We also plan to extend DDCSF to other multisource data, such as SAR and optical imagery, to further enhance its generalization capabilities.

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REFERENCES

[1] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.
[2] X. Liu, A. H.-M. Ng, F. Lei, J. Ren, L. Guo, and Z. Du, "MSLKCNN: A simple and powerful multiscale large kernel CNN for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.
[3] Y. Deng, D. Lei, Y. Peng, W. Li, and L. Zhang, "ABIBE: Adaptive building information-based extraction from remote sensing imagery using vision–language models," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.
[4] R. Li, S. Zheng, C. Duan, J. Su, and C. Zhang, "Multistage attention ResU-Net for semantic segmentation of fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[5] J. Zhuang, W. Chen, B. Guo, and Y. Yan, "Infrared weak target detection in dual images and dual areas," *Remote Sens.*, vol. 16, no. 19, Sep. 2024, Art. no. 3608, doi: 10.3390/rs16193608.
[6] G. Zhou et al., "Adaptive high-speed echo data acquisition method for bathymetric LiDAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5703017, doi: 10.1109/TGRS.2024.3386687.
[7] J. Chen, Q. Wang, W. Peng, H. Xu, X. Li, and W. Xu, "Disparity-based multiscale fusion network for transportation detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 18855–18863, Oct. 2022, doi: 10.1109/TITS.2022.3161977.
[8] W. Huang, L. Zhang, Z. Wang, and Y. Wang, "GapMatch: Bridging instance and model perturbations for enhanced semi-supervised medical image segmentation," in *Proc. AAAI Press*, 2025, pp. 17458–17466.
[9] W. Wang, L. Li, and H. Liu, "Multitarget workpiece geometric dimension measurement system based on image analysis," *Adv. Mech. Eng.*, vol. 18, no. 4, pp. 1–12, Apr. 2026, doi: 10.1177/16878132261445705.
[10] W. Ma, O. Karakuş, and P. L. Rosin, "DiverseNet: Decision diversified semi-supervised semantic segmentation networks for remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 14258–14271, 2025.
[11] B. Tu, T. Zhou, B. Liu, Y. He, J. Li, and A. Plaza, "Multi-scale autoencoder suppression strategy for hyperspectral image anomaly detection," *IEEE Trans. Image Process.*, vol. 34, pp. 5115–5130, 2025, doi: 10.1109/TIP.2025.3595408.
[12] M. Zhu, Z. Xu, Q. Zhang, Y. Liu, D. Gu, and S. S.-D. Xu, "GCSTormer: Gated Swin transformer with channel weights for image denoising," *Expert Syst. Appl.*, vol. 284, Jul. 2025, Art. no. 127924, doi: 10.1016/j.eswa.2025.127924.
[13] G. Zhou et al., "Adaptive adjustment for laser energy and PMT gain through self-feedback of echo data in bathymetric LiDAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4206422, doi: 10.1109/TGRS.2024.3403895.
[14] N. A. Almujally et al., "Multi-modal remote perception learning for object sensory data," *Frontiers Neurorobotics*, vol. 18, Sep. 2024, Art. no. 1427786, doi: 10.3389/fnbot.2024.1427786.
[15] L. Li, L. Liu, F. Cheng, Y. He, and Z. Zhong, "CN-UNet: ConvNeXt UNet with slicing-aided hyper segmentation for infrared small target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 84–98, 2026, doi: 10.1109/JSTARS.2025.3631384.
[16] Y. Dai, X.-M. Li, and H. Yuan, "Pan-Arctic winter sea ice classification using Sentinel-1 dual-polarized SAR images," *Remote Sens. Environ.*, vol. 333, Jan. 2026, Art. no. 115140, doi: 10.1016/j.rse.2025.115140.

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[17] F. Liu, J. Zhou, Z. Wang, and Z. Xu, "A three-step data acquisition optimization algorithm for hourly 3-D deformation field construction based on GNSS-InBSAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5800316, doi: 10.1109/TGRS.2026.3654658.

[18] T. Xie et al., "Iceberg detection from global open ocean Sentinel-1 wave mode SAR data with YOLO deep learning," *Remote Sens. Environ.*, vol. 337, May 2026, Art. no. 115344, doi: 10.1016/j.rse.2026.115344.

[19] C. Wang, X.-M. Li, L. Shi, H. Li, A. Mouche, and B. Chapron, "Sentinel-1 wave mode SAR monitoring of icebergs around Antarctica," *Remote Sens. Environ.*, vol. 332, Jan. 2026, Art. no. 115094, doi: 10.1016/j.rse.2025.115094.

[20] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[21] Z. Liang, S. Bao, W. Zhang, H. Yan, B. Duan, and H. Wang, "Super-resolution reconstruction of SMOS sea surface salinity from multivariate satellite observations based on deep learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 24251–24266, 2025, doi: 10.1109/JSTARS.2025.3602684.

[22] J. Liu, G. Jiang, C. Chu, Y. Li, Z. Wang, and S. Hu, "A formal model for multiagent Q-learning on graphs," *Sci. China Inf. Sci.*, vol. 68, no. 9, Sep. 2025, Art. no. 192206, doi: 10.1007/s11432-024-4289-6.

[23] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.

[24] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, *Encoder–decoder with atrous separable convolution for semantic image segmentation*, in *Computer Vision—ECCV* 2018, V. Ferrari, M. Hebert, C. Sminchisescu, and Y. Weiss, Eds., Cham, Switzerland: Springer, 2018, pp. 833–851.

[25] R. Li, S. Zheng, C. Zhang, C. Duan, L. Wang, and P. M. Atkinson, "ABCNet: Attentive bilateral contextual network for efficient semantic segmentation of fine-resolution remotely sensed imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 181, pp. 84–98, Nov. 2021.

[26] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3141–3149.

[27] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[28] H. Wu, P. Huang, M. Zhang, W. Tang, and X. Yu, "CMTFNet: CNN and multiscale transformer fusion network for remote-sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2004612.

[29] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

[30] R. Wu, X. Lu, Z. Yao, and Y. Ma, "MFMSNet: A multi-frequency and multi-scale interactive CNN-transformer hybrid network for breast ultrasound image segmentation," *Comput. Biol. Med.*, vol. 177, Jul. 2024, Art. no. 108616.

[31] X. Li et al., "Position-aware differential denoising transformer for semantic segmentation of remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 23, pp. 1–5, 2026.

[32] X. Li, F. Xu, F. Liu, Y. Tong, X. Lyu, and J. Zhou, "Semantic segmentation of remote sensing images by interactive representation refinement and geometric prior-guided inference," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[33] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. 35th Int. Conf. Neural Inf. Process. Syst. (NIPS)*, 2021, pp. 12077–12090.

[34] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[35] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[36] X. Du et al., "DP-TRAE: A dual-phase merging transferable reversible adversarial example for image privacy protection," *IEEE Trans. Dependable Secure Comput.*, vol. 22, no. 6, pp. 7849–7861, Nov. 2025, doi: 10.1109/TDSC.2025.3601175.

[37] M.-L. Zhu, J.-H. Yuan, E. Kong, L.-L. Zhao, L. Xiao, and D.-B. Gu, "Generative adversarial networks with noise optimization and pyramid coordinate attention for robust image denoising," *Int. J. Intell. Syst.*, vol. 2025, no. 1, Jan. 2025, Art. no. 1546016, doi: 10.1155/int/1546016.

[38] W. Zhang, W. Fan, W. Su, and N. Bouguila, "HKANLP: Link prediction with hyperspherical embeddings and Kolmogorov–Arnold networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 37, no. 2, pp. 966–980, Feb. 2026, doi: 10.1109/TNNLS.2025.3614341.

[39] J. Lu, J. Chen, H. Zhu, Z. Xiao, G. Xiao, and Q. Wang, "Path-based knowledge graph link prediction method with graph context," *IEEE Trans. Computat. Social Syst.*, vol. 13, no. 3, pp. 3133–3146, Jun. 2026, doi: 10.1109/TCSS.2026.3669923.

[40] X. Gan, "GraphService: Topology-aware constructor for large-scale graph applications," *ACM Trans. Archit. Code Optim.*, vol. 22, no. 1, pp. 1–24, Mar. 2025, doi: 10.1145/3689341.

[41] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018.

[42] L. Yu, Y. Li, S. Weng, H. Tian, and J. Liu, "Adaptive multi-teacher softened relational knowledge distillation framework for payload mismatch in image steganalysis," *J. Vis. Commun. Image Represent.*, vol. 95, Sep. 2023, Art. no. 103900, doi: 10.1016/j.jvcir.2023.103900.

[43] L. Yang, L. Qi, L. Feng, W. Zhang, and Y. Shi, "Revisiting weak-to-strong consistency in semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7236–7246.

[44] K. Sohn et al., "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, Eds., 2020, pp. 596–608.

[45] X. Chen, Y. Yuan, G. Zeng, and J. Wang, "Semi-supervised semantic segmentation with cross pseudo supervision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2613–2622.

[46] S. Wang, X. Sun, C. Chen, D. Hong, and J. Han, "Semi-supervised semantic segmentation for remote sensing images via multiscale uncertainty consistency and Cross-Teacher–Student attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.

[47] W. Huang, Y. Shi, Z. Xiong, and X. X. Zhu, "Decouple and weight semi-supervised semantic segmentation of remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 212, pp. 13–26, Jun. 2024.

[48] Y. Jiang et al., "SGCT-Net: SAM-guided cross-teaching network for weakly supervised semantic segmentation for generating high-quality CAMs in high-resolution remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 13395–13407, 2026, doi: 10.1109/JSTARS.2026.3682076.

[49] Y. Zhang, Z. Jiang, and H. Zhang, "ATSG: Adaptive token linking with segment anything model guidance for weakly supervised remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, doi: 10.1109/TGRS.2026.3653675.

[50] L. Wu et al., "Modeling the label distributions for weakly-supervised semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 8, pp. 6290–6306, Aug. 2025, doi: 10.1109/TPAMI.2025.3557047.

[51] G. Zhou et al., "Internal stray light suppression for single-band bathymetric LiDAR optical system," *IEEE Trans. Instrum. Meas.*, vol. 73, 2024, Art. no. 7007112, doi: 10.1109/TIM.2024.3441020.

[52] Z. Gao et al., "Exploring tourist spatiotemporal behavior differences and tourism infrastructure supply–demand pattern fusing social media and nighttime light remote sensing data," *Int. J. Digit. Earth*, vol. 17, no. 1, Dec. 2024, Art. no. 2310723, doi: 10.1080/17538947.2024.2310723.

[53] F. Wang, C. Zhang, P. Liu, B. Xie, Z. Chen, and W. Hu, "DABF-Net: A dual-branch attention-guided and bi-directional feature enhancement network for infrared small-target detection with air-to-ground benchmark," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, doi: 10.1109/TGRS.2025.3586362.

[54] A. Wei, S. Liu, C. Zhang, S. Liu, K. Xu, and Z. Chen, "FSINet: A robust feature separation and integration network for multiscale SAR object detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 8224–8238, 2026, doi: 10.1109/JSTARS.2026.3665718.

[55] A. Lin, B. Chen, J. Xu, Z. Zhang, G. Lu, and D. Zhang, "DS-TransUNet: Dual Swin transformer U-Net for medical image segmentation," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–15, 2022.

[56] S. Zhao, X. Zhang, P. Xiao, and G. He, "Exchanging dual-encoder–decoder: A new strategy for change detection with semantic guidance and spatial localization," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[57] Y. Wang, L. Yang, T. X. Wu, K. Tang, and W. Zha, "Spatial–spectral fusion BiFormer: A novel dynamic routing approach for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–15, 2025.

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[58] L. Chen, Y. Wang, and C. Zhang, "Spatial–spectral attention pyramid network for hyperspectral stripe restoration," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.
[59] X. Ma, X. Zhang, M.-O. Pun, and B. Huang, "A unified framework with multimodal fine-tuning for remote sensing semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.
[60] Y. Fu, H. Li, P. Yu, Y. Huang, and W. Zeng, "DFDR-NLNet: A dual-frequency differentiated representation non-local network for photovoltaic panel segmentation," *Appl. Energy*, vol. 401, Dec. 2025, Art. no. 126761, doi: 10.1016/j.apenergy.2025.126761.
[61] J. Wang, P. Zhang, X. Ba, and J. Chen, "Experimental and numerical investigation of segment response to seismic wave propagation for advanced geological detection in shield tunnels," *Rock Mech. Rock Eng.*, Mar. 2026, doi: 10.1007/s00603-026-05418-x.
[62] Y. Yang et al., "PestScope: Exclusion-aware large multimodal model for fine-grained agricultural pest segmentation," *IEEE Trans. Image Process.*, vol. 35, pp. 2034–2049, 2026, doi: 10.1109/TIP.2026.3661417.
[63] X. Li et al., "Frequency-guided denoising network for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026.
[64] X. Li et al., "Frequency domain-enhanced spectral–spatial fusion transformer for semantic segmentation of remote sensing images," *Inf. Fusion*, vol. 132, Aug. 2026, Art. no. 104248.
[65] X. Li, F. Xu, A. Yu, X. Lyu, H. Gao, and J. Zhou, "A frequency decoupling network for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.
[66] W. Tu et al., "Microseismic source localization method based on ABC-CS improved fusion algorithm," *Acta Geophysica*, vol. 74, no. 2, Mar. 2026, Art. no. 121, doi: 10.1007/s11600-026-01843-2.
[67] J. Yang, G. Wang, X. Xiao, M. Bao, and G. Tian, "Explainable ensemble learning method for OCT detection with transfer learning," *PLoS ONE*, vol. 19, no. 3, Mar. 2024, Art. no. e0296175, doi: 10.1371/journal.pone.0296175.
[68] X. Gan, "TianheWare: Degree-aware sampling for large-scale graph learning," *IEEE Trans. Parallel Distrib. Syst.*, vol. 37, no. 7, pp. 1–12, Jul. 2026, doi: 10.1109/TPDS.2026.3681468.
[69] S. Yun, D. Han, S. Chun, S. J. Oh, Y. Yoo, and J. Choe, "CutMix: Regularization strategy to train strong classifiers with localizable features," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6022–6031.
[70] Y. Ouali, C. Hudelot, and M. Tami, "Semi-supervised semantic segmentation with cross-consistency training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12671–12681.
[71] X. Lu et al., "Simple and efficient: A semisupervised learning framework for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.
[72] Z. Wang, Z. Zhao, X. Xing, D. Xu, X. Kong, and L. Zhou, "Conflict-based cross-view consistency for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 19585–19595.
[73] G. Liu et al., "Three-dimensional (3D) laser scanning–based identification of rock mass discontinuities for rockfall modeling using 3D discontinuous deformation analysis," *Int. J. Rock Mech. Mining Sci.*, vol. 202, Jun. 2026, Art. no. 106484, doi: 10.1016/j.ijrmms.2026.106484.
[74] B. Q. Huang and X.-M. Li, "Wave attenuation by sea ice in the Arctic marginal ice zone observed by spaceborne SAR," *Geophys. Res. Lett.*, vol. 50, no. 21, Nov. 2023, Art. no. e2023GL105059, doi: 10.1029/2023gl105059.
[75] (2010). ISPRS Potsdam. [Online]. Available: https://paperswithcode.com/dataset/isprs-potsdam
[76] X.-Y. Tong et al., "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322.
[77] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, Jun. 2010.
[78] I. Sutskever, J. Martens, G. Dahl, and G. Hinton, "On the importance of initialization and momentum in deep learning," in *Proc. Int. Conf. Mach. Learn.*, Atlanta, GA, USA, 2013, pp. 1139–1147.
[79] Y. Yang and X. Liu, "A re-examination of text categorization methods," in *Proc. 22nd Annu. Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, New York, NY, USA, Aug. 1999, pp. 42–49.
[80] Y. Yang, M. Li, S. Huang, H. Lu, W. Tu, and W. Wan, "Multi-scale spatial–spectral attention guided fusion network for pansharpening," in *Proc. 31st ACM Int. Conf. Multimedia*, 2023, pp. 3346–3354.
[81] X. Zhang, W. Huang, Q. Wang, and X. Li, "SSR-NET: Spatial–spectral reconstruction network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5953–5965, Jul. 2021.

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